An area law for four-dimensional $SU(2)$ lattice gauge theory up to $\beta \approx 0.813$

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Abstract

We consider the $SU(2)$ Wilson lattice gauge theory on the periodic torus $(\mathbb{Z}/L\mathbb{Z})^4$, with action $\beta\sum_p(1-\frac{1}{2}\operatorname{Re}\operatorname{Tr}U_p)$. We prove that for $0 \le \beta \le 0.81348$ every planar rectangular Wilson loop with sides $R \le L/2$ and $T \le L-1$ satisfies $|\langle W\rangle| \le C^T\theta^{RT}$, with explicit constants $\theta < 1$ and C independent of L; in particular the string tension is positive. The proof follows the method of Borgs and Seiler (1983, Appendix B): the slab decomposition of Durhuus and Fröhlich, a random-walk expansion of the slab σ-models and the positivity of the transfer matrix, which yields the range $\beta < 12/17 \approx 0.7059$. We refine it in two ways. (I) The ratio of slab partition functions is controlled by a strict operator inequality $\mathcal{T}^{(\lambda)} \le c(\lambda)\,\mathcal{T}$, where $c(\lambda) < 1$ comes from the Peter–Weyl spectrum (the spin-0 mode is extremal), followed by a site-by-site Cauchy–Schwarz factorisation. (II) The resulting weighted walks are counted with a memory of 11 steps, through a transfer matrix on 7 821 056 symmetry classes whose spectral radius is bounded by an integer Perron certificate. This refines the threshold of the Borgs–Seiler method from 12/17 to 0.81348 (+15.2%), with bounds for each rectangle on tori. Every numerical inequality of the proof is verified in exact integer or rational arithmetic, and the certificate is re-verified by two further programs, one of them written separately for an adversarial verification. With memory 13 the same argument gives $\beta \le 0.81498$. The result is computer-assisted, holds on tori only, and has been checked by machine (independent programs, reviews by language models, and a partial formalisation in Lean that assumes the analytic reduction), not by a human referee. The value 0.81348 lies between the estimate 4/5 announced without proof by Fröhlich (1980), in the reading discussed in §8.2, and the value 0.8222 given by the non-backtracking heuristic $5\,I_2(\beta)/I_1(\beta) < 1$, which we evaluate in §8.2. Version 1.1 adds a tilted form of the walk bound (Theorem 5.6): with the same threshold and five further integer certificates, the certified lower bound on the string tension at $\beta = 0.81348$ rises from $7.05\cdot10^{-6}$ to 0.004989, and to 0.2253 for $\beta \le 0.80$, with explicit constants on every torus.

Status labels. [proved]: complete argument written in this note, or in the cited source at the location given; [certified]: inequality between explicit numbers, verified by a program in exact integer or rational arithmetic and re-verified by at least one other program (§6); [numerical]: floating-point or extrapolated value, never used in a proof; [read]: statement read in the cited source, location given.

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1 Introduction

1.1 Context

Wilson's criterion [18] expresses the confinement of static quarks by the area law for Wilson loops. For lattice gauge theories with a compact gauge group, the area law at strong coupling

was proved by Osterwalder and Seiler [14, Thm. 5.1] by a cluster expansion, in a range of small β that is not made explicit. For the four-dimensional $SU(2)$ theory, explicit ranges come from the decomposition of Durhuus and Fröhlich [6]: conditionally on the horizontal links, the vertical links of each time slab form a three-dimensional $O(4)$ σ-model with random twists, and a rectangular Wilson loop factorises over the slabs. Table 1 lists the explicit ranges we know, in the normalisation (2.1) below.

range of β	setting	source	status
small, not explicit	planar loops; cluster expansion	[14, Thm. 5.1]	proved
$\lesssim 1/8$	refined cluster expansion of Guerra et al., as quoted in [2]	[2, p. 343]	quoted
$< 1/6$	tori, uniformly in L	[4, Thm. 1.6]	proved
$< 1/6$	written argument of [6], after correction of a factor 2 [a]	[6], [16, §7.2]	proved
≈ 0.2217	comparison with $\mathbb{Z}_2$ and Ising slabs [8] [b]	[16, §7.2]	not checked
$< 2/3$	tori, free boundary, every Gibbs measure; one-site Dobrushin	[16, Thm. 7.1]	proved
$< 12/17 \approx 0.7059$	Polyakov loops at nonzero temperature, uniformly in L_0, J_M [c]	[2, (II.55), p. 343; App. B, pp. 373–377]	proved
$< 4/5$ [d]	averaged ("in mean") estimate, "only a rough estimate"	[9, 10]	heuristic
≤ 0.81348	tori, uniformly in L; memory 11	Theorem 2.1	proved, computer-assisted
≤ 0.81498	same; memory 13	Remark 2.3	proved, computer-assisted
< 0.8222	non-backtracking heuristic $5u(\beta) < 1$	§8.2	heuristic
all β (claim)	unpublished preprint; gaps discussed in [11]	[17]	contested

Table 1: Explicit ranges of the area law for four-dimensional $SU(2)$ in the Wilson normalisation $\beta = 4/g^2$. [a] The literal statement of [6] is $\beta < 1/3$, and $\beta < 2/3$ is announced there without proof. [b] Combined with the sharpness of the three-dimensional Ising transition and $K_c^{\mathrm{Ising}}(\mathbb{Z}^3) = 0.221654626(5)$ [7], as described in [16, §7.2]; we have not seen it written out. [c] See Remark 3.7; the same method gives rectangular Wilson loops at zero temperature (§3). [d] In the reading of §8.2; the literal reading gives $8/5$.

Two observations frame this note. First, an argument that proves exponential decay of the slab correlations *uniformly in the twists* cannot go beyond the critical coupling of the untwisted $O(4)$ model on $\mathbb{Z}^3$, $\beta_c \approx 0.936$ according to Monte Carlo estimates [13, 3], since the twists may be trivial; the one-site Dobrushin criterion stops at $2/3$ exactly [16, Remark 7.6]. Second, Borgs and Seiler [2, Appendix B] do not need uniformity in the twists: after a random-walk expansion of the slab correlations, the average over the horizontal links is taken inside a trace of products of transfer matrices, and only the positivity of the transfer matrix and a bound on a ratio of slab partition functions are used. Their bound on that ratio is 1 (monotonicity in the coupling), and their count of walks only distinguishes an immediate return; the result is $J_E < (d - \frac{1}{6})^{-1}$, that is $\beta < 12/17$ for $d = 3$ space dimensions [2, (II.55), p. 343, and p. 376] [read]. They state it for correlations of Polyakov loops at nonzero temperature (Remark 3.7); the argument applies to rectangular Wilson loops at zero temperature (§3), and we call $\beta < 12/17$ the *threshold of the Borgs–Seiler method*.

1.2 Result

Theorem 2.1 proves the area law on the tori $(\mathbb{Z}/L\mathbb{Z})^4$, $L \geq 3$, uniformly in L, for $0 \leq \beta \leq 0.81348$; Remark 2.3 records $\beta \leq 0.81498$ with a larger certificate. Relative to $12/17 \approx 0.70588$ the gain is 15.2% (15.5%). It comes from two refinements of the Borgs–Seiler argument, which can be used separately or together.

(I) *A strict operator inequality* (§4). The transfer matrix $\mathcal{T}^{(\lambda)}$ of a slab with the reduced couplings produced by the walk expansion satisfies $\mathcal{T}^{(\lambda)} \leq c(\lambda)\,\mathcal{T}$, where $c(\lambda)$ is a product over links of ratios of the spin-0 eigenvalues of the one-link convolution operators; for $J > 0$, $c(\lambda) < 1$ as soon as $\lambda \not\equiv 0$. Borgs and Seiler use $c \leq 1$. A Cauchy–Schwarz inequality turns $c(\lambda)$ into a product of factors attached to the sites of the walk, which lowers the weight of every visit.

(II) *Walks with memory* (§5). The weighted walks are counted with the visits seen in a window of the last $m = 11$ steps, through a transfer matrix on 6^{11} step sequences reduced by the 48 symmetries of $\mathbb{Z}^3$; its spectral radius is bounded by an integer Perron vector, checked in 128-bit integer arithmetic.

Separately, (I) gives about 0.782 and (II) about 0.730 [numerical]; together they give 0.81348. A monotonicity property in the coupling (Lemma 5.1) lets a single certificate cover the whole interval $[0, 0.81348]$.

Version 1.1 adds a tilted form of the walk bound (§5.5). Weighting each step of the walks by $e^{\eta\langle \hat{e}, d\rangle}$, where $\hat{e}$ is the direction of the side of length R, and using the parity of walks in $\mathbb{Z}^3$, Theorem 5.6 keeps the threshold and, with five further memory-11 certificates, bounds the string tension below by 0.004989 at β_* instead of $7.05 \cdot 10^{-6}$, and by 0.2253 for $\beta \leq 0.80$ (Table 4); at β_* the bound becomes non-trivial from $R \approx 3.1 \cdot 10^3$ on every torus ($2.8 \cdot 10^3$ in the weak limits) instead of $1.9 \cdot 10^6$.

1.3 What is new and what is not

To our knowledge, the widest explicit confinement range already published for the $SU(2)$ Wilson theory in four dimensions is $\beta < 12/17 \approx 0.7059$, stated by Borgs and Seiler [2, (II.55), p. 343] and proved in their Appendix B (pp. 373–377). It concerns correlations of Polyakov loops at every temperature and every magnetic coupling; Wilson's string tension is controlled through [2, (II.50), p. 341], and the authors do not state a bound for each rectangular loop. Theorem 2.1 is a refinement of the threshold of the Borgs–Seiler method, from 12/17 to 0.81348 (+15.2%), with bounds for each rectangle on tori, uniform in the volume. It is not an independent method: the new ingredients are the strict inequality (I) with its site-by-site factorisation, its combination with finite-memory counting, and the certificate.

Not new: the slab decomposition [6]; the random-walk expansion of the slabs and the use of transfer-matrix positivity [2], which we follow closely and write out with the details we need (§3); and the bounding of walk sums by the largest eigenvalue of a finite matrix built from walks with finite memory, a classical device for the connective constant of self-avoiding walks [1, 15]. Improvement (I) is natural (it only uses the Peter–Weyl decomposition of the one-link convolution operators), and we cannot exclude that it appears in work we have not seen. Borgs and Seiler themselves point out that their bound could be sharpened by the perimeter decay of the characters, without giving a number [2, p. 377]; we do not use this route. Tomboulis [17] claims confinement in four-dimensional $SU(2)$ lattice gauge theory for all values of the coupling; this preprint has not been published, and Ito and Seiler [11] discuss gaps and open problems in its argument. We do not count it as an established range. The number itself is close to the heuristic values discussed in §8.2: it exceeds the estimate 4/5 announced without proof by Fröhlich [9, 10] in one reading of his normalisation, and it is below the value 0.8222 obtained

here from the non-backtracking heuristic, of which 4/5 is the small-β linearisation. Like every explicit strong-coupling range, the result says nothing about the weak-coupling regime relevant to the continuum limit.

1.4 Status of the proof

The analytic part (§§3–5) is proved on paper. Its numerical inputs, namely rational enclosures of twelve one-dimensional integrals and the inequality $T_{11}H \le \rho' H$ on 7 821 056 symmetry classes, are verified in exact rational and integer arithmetic and re-verified by other programs, written with independent implementation choices or in separate sessions (§6). The proof and the certificates have been checked by machine only: by independent programs, by adversarial reviews carried out by language models (see the statement on the use of AI) and, in part, in Lean; they have not been read by a human referee. The Lean formalisation (§9) covers the combinatorial part, the certificate checker and the final arithmetic; the analytic reduction of §§3–4, the monotonicity of Lemma 5.1 and the enclosures of the weights are assumed there. Theorem 5.6 (version 1.1) uses the same analytic part and five further integer certificates, each checked by two programs (§6.6); it is not formalised in Lean.

2 Setting and statement

2.1 Model and normalisation

Let $L \ge 3$ and $\Lambda_L := (\mathbb{Z}/L\mathbb{Z})^4$. Each link (z,μ), $z \in \Lambda_L$, $\mu \in \{1,2,3,4\}$, carries $U_{z,\mu} \in SU(2)$. A plaquette is $p = (z;\mu,\nu)$ with $\mu < \nu$, each geometric plaquette being counted once, and $U_p := U_{z,\mu}U_{z+e_\mu,\nu}U_{z+e_\nu,\mu}^{-1}U_{z,\nu}^{-1}$. For $\beta \ge 0$ the Wilson measure is

$$\mu_{L,\beta}(dU) = \frac{1}{Z}\exp\Big(\beta\sum_p \tfrac{1}{2}\operatorname{Re}\operatorname{Tr} U_p\Big)\prod_\ell dU_\ell, \tag{2.1}$$

with normalised Haar measure dU_ℓ; up to a constant factor the weight is $\exp(-\beta\sum_p(1-\frac{1}{2}\operatorname{Re}\operatorname{Tr}U_p))$. This is the Wilson normalisation, $\beta = 4/g^2$. We write $\langle\cdot\rangle_{L,\beta}$ for the expectation and

$$J := \beta/2 \tag{2.2}$$

for the coefficient of $\operatorname{Re}\operatorname{Tr}U_p$ in the exponent of (2.1). In other normalisations: the electric coupling J_E of Borgs and Seiler [2, (II.2)–(II.4), p. 332] at isotropic lattice spacing is $J_E = 2/g^2 = J$ [read]; the coupling of [6, (1.13)–(1.16)] and of [8] (coefficient of $\operatorname{Re}\chi$, $\chi = \operatorname{Tr}$) is J; the 't Hooft coupling of [4] is $\beta/4$.

For a closed lattice path γ, U_γ is the ordered product of its link variables and $W_\gamma := \frac{1}{2}\operatorname{Tr}U_\gamma \in [-1,1]$. A *rectangular loop with sides R along e_μ and T along e_ν* ($\mu \ne \nu$, $R,T \ge 1$) is the boundary of a planar $R\times T$ rectangle of plaquettes.

2.2 Main result

Theorem 2.1 (computer-assisted)**.** *Let*

$$J_* := \frac{40674}{100000}, \qquad \beta_* := 2J_* = 0.81348, \qquad \rho'_* := \frac{2639854083}{2^{30}},$$
$$\theta_* := J_*\rho'_* = \frac{53686712485971}{53687091200000} = 1 - 7.0540\ldots\times 10^{-6},$$

and let $C_* := K_0/(1-\theta_*)$, with K_0 defined in (5.2); numerically $K_0 = 3.97805\ldots$ and $C_* = 5.6393\ldots\times 10^5$. For every $\beta \in [0,\beta_*]$, every $L \ge 3$ and every rectangular loop γ with sides R along e_μ and T along e_ν such that $1 \le R \le L/2$ and $1 \le T \le L-1$,

$$|\langle W_\gamma\rangle_{L,\beta}| \;\le\; C_*^{\,T}\,\theta_*^{RT}. \tag{2.3}$$

Status: [proved] (§§3–5), with two [certified] numerical inputs (Proposition 5.4). The constants depend neither on L nor on $\beta \in [0, \beta_*]$. By the symmetries of the lattice the two sides may be exchanged: (2.3) with R and T exchanged holds when $T \le L/2$ and $R \le L-1$.

Corollary 2.2 (string tension)**.** *Let $0 < \beta \le \beta_*$. For all $R, T \ge 1$ and $L \ge \max(2R, T+1)$,*

$$-\frac{1}{RT} \log |\langle W_{R\times T}\rangle_{L,\beta}| \;\ge\; -\log\theta_* - \frac{\log C_*}{R}. \tag{2.4}$$

Hence

$$\sigma(\beta) := \liminf_{R\to\infty} \inf_{T\ge 1} \inf_{L\ge\max(2R,T+1)} \Big(-\frac{1}{RT} \log |\langle W_{R\times T}\rangle_{L,\beta}| \Big) \;\ge\; -\log\theta_* = 7.05\ldots\times 10^{-6} > 0,$$

and any definition of the string tension by a limit in which $R \to \infty$ (with $T \to \infty$ or not, for instance $\liminf_{R,T\to\infty}$) satisfies the same bound. The bound (2.3) also holds for every weak limit point of $(\mu_{L,\beta})_L$ as $L \to \infty$.

Proof. (2.4) is (2.3) rewritten. The limit $R \to \infty$ is needed to remove the term $(\log C_*)/R$. For a fixed loop, W_γ is a bounded continuous function of finitely many links, so (2.3) passes to weak limits. □

Remark 2.3 (memory 13). The same proof with memory $m = 13$ gives (2.3) for $\beta \le \beta_*^{(13)} := 0.81498$, with

$$J_*^{(13)} = \frac{40749}{100000}, \qquad \rho_*'^{(13)} = \frac{2635005744}{2^{30}}, \qquad \theta_*^{(13)} = \frac{6710865566391}{6710886400000} = 1 - 3.1044\ldots\times 10^{-6},$$

$K_0^{(13)} = 5.2781\ldots$ and $C_*^{(13)} = 1.7002\ldots\times 10^6$ [certified]. Its certificate has 276 294 656 symmetry classes and 4.4 GB; it is identified by its SHA-256 in §9 and can be regenerated. We state this improvement as a remark because the certificate is not part of the supplementary material.

Remark 2.4 (string-tension bounds). The same argument at $J = \beta_i/2$, with a certificate of memory 11 for each β_i, gives (2.3) for all $\beta \le \beta_i$ with the constants of Table 2 [certified]. For $\beta \le 0.66$ the bound $\sigma \ge \operatorname{arcosh}(2/\beta - 2)$ of [16, Cor. 7.2] is better (for instance 0.795 against 0.256 at $\beta = 0.6$); it tends to 0 as $\beta \uparrow 2/3$. Corollary 5.8 (version 1.1) gives $\sigma \ge 0.6080$ for all $\beta \le 0.70588$, which is better than $\operatorname{arcosh}(2/\beta - 2)$ for $\beta > 0.627$. Without the tilt, the walk method is useful for the range rather than for the size of the tension. Near the threshold the bound (2.3) is non-trivial only at large distances: $C^T\theta^{RT} < 1$ if and only if $R > R_{\min} := \log C/(-\log\theta)$. From the constants of Table 2, $R_{\min} \approx 12$ for $\beta \le 0.60$, 143 for $\beta \le 0.78$, 430 for $\beta \le 0.80$, $2.1\cdot 10^3$ for $\beta \le 0.81$ and $1.9\cdot 10^6$ for $\beta \le \beta_*$ (with $C_* = 5.6\cdot 10^5$ and $-\log\theta_* = 7.1\cdot 10^{-6}$); at β_* the bound is purely asymptotic. With Theorem 5.6 (version 1.1), Table 4 gives instead $R_{\min} \le 8.5$, 36.5, 85.5 and 3 107 on every torus (6.3, 30.3, 72.4 and 2 829 in the weak limits) for $\beta \le 0.70588$, 0.80, 0.81 and β_*.

3 The method of Borgs and Seiler

This section writes out, in our notation and for rectangular Wilson loops on tori, the argument of [2, Appendix B]. We take $\nu = 4$ (the side of length T is vertical) and $\mu = 1$; the general case follows by symmetry.

β_i	θ_i	$-\log\theta_i$	C_i
0.60	0.77398	0.2562	19.3
0.66	0.84102	0.1731	26.8
0.70588	0.89048	0.1160	38.2
0.72	0.90538	0.0994	43.9
0.74	0.92621	0.0767	55.8
0.76	0.94672	0.0547	76.6
0.78	0.96692	0.0336	122
0.80	0.98679	0.0133	303
0.81	0.99660	0.0034	$1.2\cdot10^{3}$
0.81348	0.9999929	$7.1\cdot10^{-6}$	$5.6\cdot10^{5}$

Table 2: Constants of (2.3) valid for all $\beta\le\beta_i$ (memory 11). Each row rests on its own weights and integer certificate, produced and checked as in §6 and re-verified on all 6^{11} states by an independent program; only the last one is in the supplementary material, the others can be regenerated with the programs of §9. The weights files of the rows 0.70588, 0.80 and 0.81 are in `supplement_v1.1` (version 1.1).

3.1 Slabs

Let $\iota : S^3 \to SU(2)$, $\iota(q) = \begin{pmatrix} q^0+iq^3 & -q^1+iq^2 \\ q^1+iq^2 & q^0-iq^3 \end{pmatrix}$ (the parametrisation of [6, (5.1)]); it maps the normalised surface measure $\bar\sigma$ of $S^3\subset\mathbb{R}^4$ onto Haar measure, and $\frac12\operatorname{Re}\operatorname{Tr}(\iota(q)^{-1}\iota(r)) = \langle q,r\rangle$. For $A,B\in SU(2)$ the map $r\mapsto\iota^{-1}(A\,\iota(r)B^{-1})$ is the restriction to S^3 of a matrix $O_{A,B}\in SO(4)$, with $O_{A,B}O_{A',B'}=O_{AA',BB'}$ and $O_{A,B}^{\mathsf T}=O_{A^{-1},B^{-1}}$.

Let $X:=(\mathbb{Z}/L\mathbb{Z})^3$, with its set E_X of $3L^3$ nearest-neighbour edges; for $L\ge3$ it is a simple 6-regular graph. The *vertical* links are $Q_x^k:=U_{(x,k),4}$, $x\in X$, $k\in\mathbb{Z}/L\mathbb{Z}$, and $q_x^k:=\iota^{-1}(Q_x^k)$; the other links are *horizontal*, and $\mathcal{H}$ is the σ-algebra they generate. A vertical plaquette $p=((x,k);\mu,4)$, $\mu\le3$, has $U_p = A\,Q^k_{x+e_\mu}B^{-1}(Q^k_x)^{-1}$ with $A=U_{(x,k),\mu}$ (layer k) and $B=U_{(x,k+1),\mu}$ (layer $k+1$), so that $\frac12\operatorname{Re}\operatorname{Tr}U_p=\langle q_x^k, O_{A,B}\,q^k_{x+e_\mu}\rangle$.

Lemma 3.1 (slabs; [6, (1.17)–(1.19)], [16, Lemma 7.4])**.** *Conditionally on $\mathcal{H}$, the families $(q_x^k)_{x\in X}$, $k\in\mathbb{Z}/L\mathbb{Z}$, are independent, and $(q_x^k)_x$ has the law*

$$\mu_k(dq)=\frac{1}{Z_k}\exp\Big(2J\sum_{x\in X,\,\mu\le3}\langle q_x, O^k_{x,x+e_\mu}\,q_{x+e_\mu}\rangle\Big)\prod_{x\in X}\bar\sigma(dq_x), \tag{3.1}$$

with $O^k_{x,x+e_\mu}:=O_{U_{(x,k),\mu},U_{(x,k+1),\mu}}$.

Proof. A horizontal plaquette contains no vertical link, and a vertical plaquette contains two vertical links of the same slab; the density (2.1) therefore factorises over the slabs given $\mathcal{H}$, and $\beta\cdot\frac12\operatorname{Re}\operatorname{Tr}U_p = 2J\langle q_x, O\,q_{x+e_\mu}\rangle$. □

We write $O_{yx}:=O_{xy}^{\mathsf T}$ for the reversed edge, and $Z_k(\boldsymbol{J})$ for the normalisation of (3.1) with the coupling J of each edge ℓ replaced by J_ℓ; it is a function of the horizontal links of layers k and $k+1$.

3.2 Random-walk expansion of a slab

Lemma 3.2 (integration by parts on a sphere)**.** *For $F\in C^1(\mathbb{R}^4)$, $c>0$ and $a\in\{0,1,2,3\}$,* $\displaystyle\int s_aF(s)\,\delta(|s|^2-c)\,d^4s=\frac12\int_0^c dc'\int\partial_aF(s)\,\delta(|s|^2-c')\,d^4s.$

Proof. Let $\Theta(c) := \int_{|s|^2\le c} \partial_a F\, d^4s$. By the divergence theorem $\Theta(c) = \int_{|s|=\sqrt{c}} F\, s_a/\sqrt{c}\, dS$; by the coarea formula $\int g\,\delta(|s|^2-c)\,d^4s = (2\sqrt{c})^{-1}\int_{|s|=\sqrt{c}} g\, dS$ and $\Theta'(c) = \int \partial_a F\,\delta(|s|^2-c)\,d^4s$. Hence the left side equals $\Theta(c)/2 = \frac{1}{2}\int_0^c \Theta'$. □

The factor $\frac{1}{2}$ does not appear in the printed form of [2, Lemma B.1(i), p. 374]; it is compensated elsewhere in their computation, and the net weight per step is J in both (an independent check of the normalisation is in §7).

Let G be a finite simple graph, $J \ge 0$, $O_{xy} \in SO(4)$ for each ordered pair of neighbours with $O_{yx} = O_{xy}^{\mathsf{T}}$, and

$$Z(\boldsymbol{J}) := \int \exp\Big(2 \sum_{\{u,v\}\in E_G} J_{uv}\langle q_u, O_{uv} q_v\rangle\Big) \prod_{z\in G} \bar\sigma(dq_z) \qquad (\boldsymbol{J} = (J_{uv})_{\{u,v\}\in E_G}),$$

and let μ be the corresponding probability measure for the constant couplings $J_{uv} \equiv J$. A *walk* $\omega : x \to y$ is a sequence $\omega_0 = x, \omega_1, \dots, \omega_{|\omega|} = y$ of sites with ω_{i+1} a neighbour of ω_i; returns and self-intersections are allowed. Put $O_\omega := O_{\omega_0\omega_1}\cdots O_{\omega_{|\omega|-1}\omega_{|\omega|}}$, $n_z(\omega) := \#\{i : \omega_i = z\}$, $V(\omega) := \{z : n_z(\omega) \ge 1\}$,

$$f_n(\lambda) := \frac{\lambda^{n-1}(1-\lambda)}{(n-1)!} \quad (n \ge 1,\ \lambda \in [0,1]), \qquad F_\omega(\lambda) := \prod_{z\in V(\omega)} f_{n_z(\omega)}(\lambda_z), \tag{3.2}$$

and, for $\lambda \in [0,1]^{V(\omega)}$ extended by $\lambda_u := 0$ for $u \notin V(\omega)$, the reduced couplings $J_{uv}^\lambda := J\sqrt{(1-\lambda_u)(1-\lambda_v)}$.

Proposition 3.3 (walk representation; [2, (B.1)–(B.8), pp. 374–375])**.** *For $x \ne y$ in G and $a, b \in \{0,\dots,3\}$,*

$$\mathbb{E}_\mu[q_{x,a}\, q_{y,b}] = \frac{1}{2}\sum_{\omega: x\to y} J^{|\omega|}\,(O_\omega)_{ab} \int_{[0,1]^{V(\omega)}} F_\omega(\lambda)\,\frac{Z(\boldsymbol{J}^\lambda)}{Z(J)}\,d\lambda, \tag{3.3}$$

and the series converges absolutely.

Proof. Write $\bar\sigma(dq) \propto \delta(|s|^2-1)\,d^4s$ (the constant cancels in the ratio) and e^H for the weight, so that $\partial_{s_z,c} H = 2J\sum_{u\sim z}(O_{zu}s_u)_c$. (i) Apply Lemma 3.2 to the factor $s_{x,a}$: since $\frac{1}{2}\partial_{x,a} e^H = J\sum_{u\sim x}(O_{xu}s_u)_a e^H$, this produces a step to a neighbour u with weight $\frac{1}{2}\cdot 2J = J$, a new linear factor $(O_{xu}s_u)_a$, and replaces the constraint at x by $\delta(|s_x|^2 - (1-\lambda_x))$ with $\lambda_x \in [0,1]$. (ii) Repeat with the linear factor at the current site. When the current site is y, the derivative may also hit $s_{y,b}$; this gives $\frac{1}{2}\delta_{cb}$ and ends the walk, which is the global factor $\frac{1}{2}$ of (3.3). The derivative of H always carries the unreduced coupling J. (iii) At the n-th visit of a site z the new parameter runs over $[\lambda_z^{(n-1)}, 1]$ and the integrand depends only on the last one; the ordered simplex gives $\lambda^{n-1}/(n-1)!$. (iv) At the end, the change of variables $s_z = \sqrt{1-\lambda_z}\,s'_z$ gives $\delta(|s_z|^2-(1-\lambda_z))\,d^4s_z = (1-\lambda_z)\,\delta(|s'_z|^2-1)\,d^4s'_z$ (this is where the dimension 4 of $\mathbb{R}^4$ enters), and $2J\langle s_u, O\,s_v\rangle = 2J^\lambda_{uv}\langle s'_u, O\,s'_v\rangle$; hence the factor $Z(\boldsymbol{J}^\lambda)/Z(J)$. (v) Convergence: after N applications of Lemma 3.2 the identity holds with a remainder that is a finite sum over the unfinished walks of length N. Each term still contains a linear factor of absolute value at most 1 (all radii are at most 1) and $s_{y,b}$, and is bounded by $e^{4J|E_G|}J^N\prod_z 1/(n_z-1)!$. On a graph with $|G|$ sites, $\sum_z(n_z-1) \ge N+1-|G|$, so $\prod_z(n_z-1)!$ grows faster than any exponential of N, while there are at most Δ_G^N walks of length N (Δ_G the maximal degree). The remainder tends to 0, and the finished terms are dominated in the same way. □

Corollary 3.4 (matrix elements)**.** *In slab k, for $x \ne y$, conditionally on $\mathcal{H}$,*

$$\mathbb{E}\big[(Q_x^k)_{ij}\,((Q_y^k)^{-1})_{lk'}\,\big|\,\mathcal{H}\big] = \sum_{\omega: x\to y} J^{|\omega|}\,(A_\omega)_{ik'}\,(B_\omega^{-1})_{lj}\int F_\omega(\lambda)\,\frac{Z_k(\boldsymbol{J}^\lambda)}{Z_k(J)}\,d\lambda,$$

where A_ω (resp. B_ω) is the ordered product of the horizontal links along ω in layer k (resp. $k+1$).

Proof. Write $\iota(q) = \sum_a q_a e_a$ with $e_a \in M_2(\mathbb{C})$. Then

$$\sum_a (e_a)_{pq}\overline{(e_a)_{rs}} = 2\delta_{pr}\delta_{qs}, \qquad \sum_a (O_{A,B})_{ab} e_a = A\, e_b B^{-1},$$

so $O_\omega = O_{A_\omega, B_\omega}$ and $\sum_{a,b}(O_\omega)_{ab}(e_a)_{ij}\overline{(e_b)_{k'l}} = 2(A_\omega)_{ik'}(B_\omega^{-1})_{lj}$. Insert (3.3); the factor 2 cancels the $\frac{1}{2}$. □

3.3 The Wilson loop as a sum over sequences of walks

Let γ have corners $(x,0)$, $(y,0)$, (y,T), (x,T) with $y = x + Re_1$; let A be the product of the R horizontal links from x to y in layer 0 and B that from y to x in layer T. Then $U_\gamma = A\, Q_y^0 \cdots Q_y^{T-1}\, B\, (Q_x^{T-1})^{-1} \cdots (Q_x^0)^{-1}$.

Proposition 3.5. *With walks $\omega_k : y \to x$ in X, one for each slab $k = 0, \dots, T-1$,*

$$\langle W_\gamma \rangle_{L,\beta} = \frac{1}{2} \sum_{\omega_0,\dots,\omega_{T-1}} \prod_{k<T} J^{|\omega_k|} \int \prod_{k<T} F_{\omega_k}(\lambda^{(k)})\, d\lambda^{(k)}\ \mathbb{E}\Big[\prod_{k<T} \frac{Z_k(\boldsymbol{J}^{\lambda^{(k)}})}{Z_k(J)} \prod_{m=0}^{T} \operatorname{Tr} U_{C_m} \Big], \tag{3.4}$$

where $\mathbb{E}$ is the expectation of the horizontal links under $\mu_{L,\beta}$ and C_m is a closed loop of layer m: C_0 is the bottom side ($x \to y$) followed by ω_0 ($y \to x$); for $1 \le m \le T-1$, C_m is ω_{m-1} reversed ($x \to y$) followed by ω_m; C_T is ω_{T-1} reversed followed by the top side ($y \to x$).

Proof. Contraction. For matrices P, N that do not depend on the vertical links of slab k, Corollary 3.4 (walks from y to x) gives, conditionally on $\mathcal{H}$ and on the other slabs,

$$\mathbb{E}_k\big[\operatorname{Tr}\,(P\, Q_y^k N\, (Q_x^k)^{-1})\big] = \sum_\omega w_\omega\ \operatorname{Tr}(P A'_\omega)\ \operatorname{Tr}(B'^{-1}_\omega N),$$

with w_ω the weight of ω and A'_ω, B'_ω its holonomies from y to x in layers k and $k+1$: expand $\operatorname{Tr}(PQ_yNQ_x^{-1}) = \sum P_{ai}(Q_y)_{ij}N_{jl}(Q_x^{-1})_{la}$ and replace $(Q_y)_{ij}(Q_x^{-1})_{la}$ by $(A'_\omega)_{ia}(B'^{-1}_\omega)_{lj}$. Apply this to slab 0 with $P = A$, which produces $\operatorname{Tr} U_{C_0}$ and leaves an expression of the same form with $P = B'^{-1}_{\omega_0}$; by induction over the slabs (independent given $\mathcal{H}$, Lemma 3.1) one obtains $\prod_{m=0}^T \operatorname{Tr} U_{C_m}$. *Fubini.* For all λ and all horizontal configurations, $0 < Z_k(\boldsymbol{J}^\lambda)/Z_k(J) \le e^{12JL^3}$ (each of the $3L^3$ edges changes the exponent by at most $2J$), $|F_\omega| \le \prod_z 1/(n_z-1)!$ and $|\operatorname{Tr} U_C| \le 2$; the series $\sum_\omega J^{|\omega|}\prod_z 1/(n_z-1)!$ converges (proof of Proposition 3.3). Sums, integrals and expectation may therefore be exchanged. □

3.4 Transfer matrix

Let $\mathfrak{h} := L^2(SU(2)^{E_X})$ with Haar measure. For $J \ge 0$ let t_J be the convolution operator $(t_J f)(v) = \int e^{J \operatorname{Re}\operatorname{Tr}(vw^{-1})} f(w)\, dw$ on $L^2(SU(2))$, and for $\boldsymbol{J} = (J_\ell)_{\ell \in E_X}$ let $T_0(\boldsymbol{J}) := \bigotimes_\ell t_{J_\ell}$. Let P_0 be the average over gauge transformations $(u\cdot v)_{x,\mu} = u_x v_{x,\mu} u_{x+e_\mu}^{-1}$, $u \in SU(2)^X$ (the orthogonal projection onto gauge-invariant functions), and M the multiplication by $\exp(J\sum_p \operatorname{Re}\operatorname{Tr} v_p)$, the sum running over the $3L^3$ horizontal plaquettes of a layer. Then:

- $Z_k(\boldsymbol{J})(v_k, v_{k+1}) = \int \prod_x du_x \prod_\ell e^{J_\ell \operatorname{Re}\operatorname{Tr}(v_{k,\ell}(u\cdot v_{k+1})_\ell^{-1})}$ is the kernel of $T_0(\boldsymbol{J})P_0$ (the vertical links of slab k act as a gauge transformation of layer $k+1$);
- $T_0(\boldsymbol{J})$ commutes with gauge transformations (each t_{J_ℓ} is a convolution by a class function), hence with P_0;

- M and the functions $v \mapsto \operatorname{Tr} U_C(v)$ of closed loops C are gauge invariant, so the corresponding multiplication operators commute with P_0.

By Lemma 4.1 below, $t_J \geq 0$ is trace class, and so is $T_0(\boldsymbol{J})$. We put, for $\lambda \in [0,1]^{V(\omega)}$,

$$\mathcal{T} := M^{1/2} P_0 T_0(J) P_0 M^{1/2} \ \geq 0, \qquad \mathcal{T}^{(\lambda)} := M^{1/2} P_0 T_0(\boldsymbol{J}^{\lambda}) P_0 M^{1/2} \ \geq 0.$$

Proposition 3.6 (trace formula; [2, (B.10), p. 375]). *For walks $\omega_0, \dots, \omega_{T-1}$ and parameters $\lambda^{(k)}$,*

$$\mathbb{E}\Big[\prod_{k<T} \frac{Z_k(\boldsymbol{J}^{\lambda^{(k)}})}{Z_k(J)} \prod_{m=0}^{T} \operatorname{Tr} U_{C_m} \Big] = \frac{\operatorname{Tr}\big(\prod_{m=0}^{L-1} \chi_m \mathcal{T}_m \big)}{\operatorname{Tr} \mathcal{T}^L}, \tag{3.5}$$

where χ_m is the multiplication by $\operatorname{Tr} U_{C_m}$ for $m \leq T$ and $\chi_m := 1$ for $m > T$, and $\mathcal{T}_m := \mathcal{T}^{(\lambda^{(m)})}$ for $m < T$, $\mathcal{T}_m := \mathcal{T}$ for $m \geq T$ (the product is ordered by increasing m).

Proof. Each plaquette is either horizontal (in one layer) or vertical (in one slab), and the vertical links of slab k are integrated in Z_k; hence the marginal law of the horizontal links is $Z^{-1} \prod_m M(v_m) \prod_k Z_k(J)(v_k, v_{k+1}) \prod dv$, with $Z = \operatorname{Tr}\big((M T_0(J) P_0)^L\big)$. Multiplying by the ratios replaces the kernel of slab k by that of $T_0(\boldsymbol{J}^{\lambda^{(k)}}) P_0$, and the left side of (3.5) becomes $Z^{-1} \operatorname{Tr}\big(\prod_m \chi_m M T_0(\boldsymbol{J}^{(m)}) P_0 \big)$. Write $M = M^{1/2} M^{1/2}$ and $T_0 P_0 = P_0 T_0 P_0$; since P_0, $M^{1/2}$ and the χ_m commute, cyclicity of the trace gives $\operatorname{Tr}(\prod_m \chi_m \mathcal{T}_m)$, and likewise $Z = \operatorname{Tr} \mathcal{T}^L$. □

3.5 The bound of Borgs and Seiler

Borgs and Seiler note that $\mathcal{T}^{(\lambda)} \leq \mathcal{T}$, since $\langle \varphi, \mathcal{T}^{(\lambda)} \varphi \rangle$ is non-decreasing in each coupling by reflection positivity [2, (B.11), p. 376]; Hölder's inequality then bounds the right side of (3.5) by 2^{T+1} (Proposition 4.3 below with $c \leq 1$). Since $\int_0^1 f_n = a_n := 1/(n+1)!$ [2, (B.14), p. 376], (3.4) gives

$$|\langle W_\gamma \rangle| \leq 2^T G(x,y)^T, \qquad G(x,y) := \sum_{\omega : y \to x} J^{|\omega|} \prod_{z \in V(\omega)} a_{n_z(\omega)}. \tag{3.6}$$

Read along the walk, $\prod_z a_{n_z(\omega)} = \prod_{j=0}^{|\omega|} r(N_j)$ with $N_j := \#\{i \leq j : \omega_i = \omega_j\}$, $r(1) = a_1 = \frac{1}{2}$ and $r(N) = a_N / a_{N-1} = 1/(N+1)$ for $N \geq 2$. At each step after the first, the return to the previous site has $N \geq 2$, so the six continuations contribute at most $J(5 \cdot \frac{1}{2} + \frac{1}{3}) = \frac{17}{6} J$ [2, p. 376]. Hence $G(x,y) \leq \frac{3}{2} \sum_{n \geq R} J^n (\frac{17}{6})^{n-1}$, and (3.6) gives the area law for $J < 6/17$, i.e. $\beta < 12/17$ [2, p. 376] [read].

Remark 3.7 (what Borgs and Seiler state). Borgs and Seiler state the bound for the correlation of Polyakov loops at nonzero temperature: "Irrespective of J_M and temperature we are sure to have confinement for $J_E < \tilde{J}_c$", with $\tilde{J}_c = (d - \frac{1}{6})^{-1}$ for $SU(2)$ [2, (II.55), p. 343], proved in their Appendix B (pp. 373–377), uniformly in the number L_0 of time layers; they relate the Polyakov and Wilson string tensions in [2, (II.48)–(II.50), p. 341] [read]. The version for rectangular Wilson loops at zero temperature, with constants, is the adaptation written in §§3–3.5; in this note "the threshold of the Borgs–Seiler method" refers to the resulting range $\beta < 12/17$. Page numbers of [2] refer to the journal version; the preprint MPI-PAE/PTh 16/83 has the same text and equation numbers in all the passages cited.

4 Improvement (I): a strict operator inequality

Lemma 4.1 (Peter–Weyl spectrum). *On the span of the matrix coefficients of the spin-j representation, $j \in \{0, \frac{1}{2}, 1, \dots\}$, the operator t_J acts as multiplication by*

$$m_j(J) = \frac{I_{2j+1}(2J)}{J} = \sum_{k \geq 0} \frac{J^{2k+2j}}{k!\,(k+2j+1)!} \quad (J > 0), \qquad m_0(0) = 1, \ m_j(0) = 0 \ (j \geq \tfrac{1}{2}).$$

In particular $t_J \geq 0$ and t_J is trace class. For $0 < J \leq 1$ and $0 \leq t \leq 1$, $m_j(tJ)/m_j(J) \leq m_0(tJ)/m_0(J)$ for every j.

Proof. Convolution by the class function $\varphi(g) = e^{J \operatorname{Tr} g}$ acts on the spin-j component by $c_j/(2j+1)$, $c_j := \int \varphi\, \chi_j$. With $\operatorname{Tr} g = 2\cos\vartheta$, $\chi_j = \sin((2j+1)\vartheta)/\sin\vartheta$ and Weyl's integration formula, $c_j = \frac{2}{\pi}\int_0^\pi e^{2J\cos\vartheta}\sin((2j+1)\vartheta)\sin\vartheta\, d\vartheta = I_{2j}(2J) - I_{2j+2}(2J) = (2j+1)I_{2j+1}(2J)/J$, and the series is that of I_{2j+1}. The eigenvalues are positive and $\sum_j (2j+1)^2 m_j(J) < \infty$. For $j \geq \frac{1}{2}$, termwise comparison gives $m_j(tJ) \leq t^{2j} m_j(J) \leq t\, m_j(J)$. For $j = 0$, $m_0(tJ)/m_0(J) = \sum_k w_k t^{2k}$ with probability weights $w_k \propto J^{2k}/(k!(k+1)!)$; by Jensen's inequality this is at least $t^{2\langle k\rangle}$, and $2\langle k\rangle \leq 1$ because $\sum_{k\geq 1}(2k-1)J^{2k}/(k!(k+1)!) \leq \sum_{k\geq 1}(2k-1)/(k!(k+1)!) < 0.79$ for $J \leq 1$. Hence $m_0(tJ)/m_0(J) \geq t$. (For $t = 0$ the left side vanishes for $j \geq \frac{1}{2}$ and equals $1/m_0(J)$ for $j = 0$.) □

For $s \in [0,1]$ put

$$M_J(s) := \sum_{k\geq 0} \frac{(J^2 s)^k}{k!\,(k+1)!} = m_0(J\sqrt{s}), \qquad \rho_J(s) := \frac{M_J(s)}{M_J(1)} \in (0,1]. \tag{4.1}$$

Proposition 4.2 (strict operator inequality)**.** *Let $0 < J \leq 1$, ω a walk in X and $\lambda \in [0,1]^{V(\omega)}$. Then $0 \leq \mathcal{T}^{(\lambda)} \leq c(\lambda)\,\mathcal{T}$ with*

$$c(\lambda) := \prod_{\{u,v\}\in E_X} \rho_J((1-\lambda_u)(1-\lambda_v)) \qquad (\lambda_u := 0 \text{ for } u \notin V(\omega)).$$

The constant is optimal link by link.

Proof. $T_0(\boldsymbol{J}^\lambda)$ and $T_0(J)$ are diagonal in the product Peter–Weyl basis, with eigenvalues $\prod_\ell m_{j_\ell}(J_\ell^\lambda)$ and $\prod_\ell m_{j_\ell}(J)$. Since $J_\ell^\lambda = J\sqrt{s_\ell}$ with $s_\ell = (1-\lambda_u)(1-\lambda_v)$, Lemma 4.1 gives $T_0(\boldsymbol{J}^\lambda) \leq c(\lambda) T_0(J)$; conjugation by $M^{1/2}P_0$ preserves the order. The constant function (all $j_\ell = 0$) is gauge invariant and gives equality for T_0. □

Borgs and Seiler use $\mathcal{T}^{(\lambda)} \leq \mathcal{T}$, i.e. $c \leq 1$. Here $c(\lambda) < 1$ as soon as $\lambda \not\equiv 0$. This is improvement (I). (For $J = 0$ the operators $\mathcal{T}^{(\lambda)}$ and $\mathcal{T}$ coincide, $c \equiv 1$ and $\langle W_\gamma\rangle = 0$: the case is trivial, and it is excluded from the statements above, as in §5.4.)

Proposition 4.3 (Hölder bound)**.** *In the setting of Proposition 3.6, with $c_k := c(\lambda^{(k)})$,*

$$\Big|\mathbb{E}\Big[\prod_{k<T} \frac{Z_k(\boldsymbol{J}^{\lambda^{(k)}})}{Z_k(J)} \prod_{m=0}^{T} \operatorname{Tr} U_{C_m}\Big]\Big| \leq 2^{T+1} \prod_{k<T} c_k.$$

Proof. If $0 \leq A \leq B$ are bounded operators, then $\|A^{1/2} f\| \leq \|B^{1/2} f\|$, so there is a contraction C with $A^{1/2} = CB^{1/2}$ [5] and $A = B^{1/2}C^*CB^{1/2}$. With $A = \mathcal{T}_k$ and $B = c_k\mathcal{T}$ this gives $\mathcal{T}_k = \mathcal{T}^{1/2} Y_k \mathcal{T}^{1/2}$ with $0 \leq Y_k$, $\|Y_k\| \leq c_k$ (and $Y_m = I$ for $m \geq T$). By cyclicity, $\operatorname{Tr}(\prod_m \chi_m \mathcal{T}_m) = \operatorname{Tr}(\prod_m X_m Y_m)$ with $X_m := \mathcal{T}^{1/2}\chi_m\mathcal{T}^{1/2}$. Hölder's inequality for Schatten norms, with L factors of exponent L and the Y_m in operator norm, gives $|\operatorname{Tr}(\cdots)| \leq \prod_m \|X_m\|_L \|Y_m\|$, and $\|X_m\|_L \leq \|\chi_m\|_\infty \|\mathcal{T}^{1/2}\|_{2L}^2 = \|\chi_m\|_\infty (\operatorname{Tr}\mathcal{T}^L)^{1/L}$. Conclude with $\|\chi_m\|_\infty \leq 2$ for $m \leq T$, $= 1$ otherwise, and (3.5). □

Lemma 4.4 (Cauchy–Schwarz)**.** *For $a, b \in [0,1]$: $M_J(ab)^2 \leq M_J(a^2)M_J(b^2)$ and $M_J(a) \leq M_J(a^2)^{1/2} M_J(1)^{1/2}$.*

Proof. Cauchy–Schwarz for $\sum_k c_k (a^k)(b^k)$ with $c_k = J^{2k}/(k!(k+1)!) \geq 0$; the second inequality is the case $b = 1$. □

Corollary 4.5 (site-by-site factorisation)**.** *Let $L \geq 3$, ω a walk in X and $\lambda \in [0,1]^{V(\omega)}$. Then*

$$c(\lambda) \leq \prod_{z \in V(\omega)} h_J(\lambda_z), \qquad h_J(\lambda) := \rho_J\big((1-\lambda)^2\big)^3.$$

Proof. Put $a_z := 1 - \lambda_z$. By Lemma 4.4, an edge $\{u, v\} \subset V(\omega)$ contributes $\rho(a_u a_v) \leq \rho(a_u^2)^{1/2}\rho(a_v^2)^{1/2}$, an edge $\{u, w\}$ with $w \notin V(\omega)$ contributes $\rho(a_u) \leq \rho(a_u^2)^{1/2}$, and the other edges contribute 1. Every site of X has exactly six incident edges, so every $z \in V(\omega)$ receives exactly six factors $\rho(a_z^2)^{1/2}$. □

The exponent 3 uses that every site has exactly six edges, which holds on the torus with $L \geq 3$. A site with $k < 6$ edges (for instance on the boundary of a box) would receive $\rho(a_z^2)^{k/2} \geq \rho(a_z^2)^3$; this is why the theorem is stated on tori only (§8.1).

New weights. Let

$$\widetilde{A}_n(J) := \int_0^1 f_n(\lambda)\, h_J(\lambda)\, d\lambda \;\leq\; a_n = \frac{1}{(n+1)!}, \qquad \widetilde{r}(1) := \widetilde{A}_1, \quad \widetilde{r}(N) := \frac{\widetilde{A}_N}{\widetilde{A}_{N-1}} \;(N \geq 2). \tag{4.2}$$

Combining (3.4), Proposition 4.3 and Corollary 4.5 (all integrands are non-negative after the bound), and integrating site by site,

$$|\langle W_\gamma \rangle_{L,\beta}| \leq 2^T\, \widetilde{G}(x,y)^T, \qquad \widetilde{G}(x,y) := \sum_{\omega: y \to x} J^{|\omega|} \prod_{z \in V(\omega)} \widetilde{A}_{n_z(\omega)} = \sum_{\omega: y \to x} J^{|\omega|} \prod_{j=0}^{|\omega|} \widetilde{r}(N_j), \tag{4.3}$$

with $N_j = \#\{i \leq j : \omega_i = \omega_j\}$ as in §3.5: the weight is read step by step.

Lemma 4.6 (revisits)**.** *$\widetilde{r}(N) \leq 1/(N+1)$ for $N \geq 2$.*

Proof. $\widetilde{r}(N) = \mathbb{E}_\nu[\lambda]/(N-1)$ with $\nu \propto \lambda^{N-2}(1-\lambda)h_J(\lambda)\, d\lambda$. The function h_J is non-increasing (ρ_J is non-decreasing and $(1-\lambda)^2$ decreasing). By Chebyshev's integral inequality for the positive measure $w = \lambda^{N-2}(1-\lambda)\, d\lambda$, the increasing function λ and the non-increasing function h_J, $\mathbb{E}_\nu[\lambda] = \int w\lambda h_J / \int w h_J \leq \int w\lambda / \int w = (N-1)/(N+1)$. □

Remark 4.7. With the count of §3.5 (memory 1), (I) alone gives the range $\beta \lesssim 0.7824$ [numerical]. At $J = J_*$ one has $\widetilde{A}_1 = 0.443739\ldots$ instead of $a_1 = \frac{1}{2}$.

5 Improvement (II): walks with memory, and proof of Theorem 2.1

5.1 Monotonicity in the coupling

Lemma 5.1. *(M1) For every λ, $J \mapsto h_J(\lambda)$ is non-increasing on $(0, \infty)$. (M2) For $0 < J \leq 0.44$, $J \mapsto J\, h_J(\lambda)$ is non-decreasing. Consequently, for $0 < J \leq J' \leq 0.44$ and every $N \geq 1$, $J\, \widetilde{r}(N; J) \leq J'\, \widetilde{r}(N; J')$.*

Proof. Let $F(x) := \sum_k x^k/(k!(k+1)!)$ and $\varphi(x) := xF'(x)/F(x)$, the mean of k under the weights $x^k/(k!(k+1)!)$, which is non-decreasing in x (its logarithmic derivative is a variance). Then $\partial_J \log \rho_J(s) = \frac{2}{J}(\varphi(J^2 s) - \varphi(J^2)) \in [-\frac{2}{J}\varphi(J^2), 0]$, so $\partial_J \log h_J \in [-\frac{6}{J}\varphi(J^2), 0]$, which is (M1), and $\partial_J \log(J h_J) \geq \frac{1}{J}(1 - 6\varphi(J^2))$. For $J \leq 0.44$, $J^2 \leq 0.2$ and, since $F \geq 1$, $\varphi(J^2) \leq 0.2\, F'(0.2) = \sum_{k \geq 1} k\,(0.2)^k/(k!(k+1)!) = 0.10683557\ldots < \frac{1}{6}$: the first 40 terms are summed in exact rational arithmetic, and the tail is at most $t_{41}/(1 - 0.2/(41 \cdot 43)) < 2.5 \cdot 10^{-128}$, where $t_k := k(0.2)^k/(k!(k+1)!)$ satisfies $t_{k+1}/t_k = 0.2/(k(k+2))$. This proves (M2). Hence $J\widetilde{A}_N(J) = \int f_N \cdot (J h_J)$ is non-decreasing and $\widetilde{A}_{N-1}(J)$ is non-increasing, so $J\widetilde{r}(N; J)$ is non-decreasing ($N = 1$: $J\widetilde{A}_1$). □

5.2 Counting with a finite memory

Let $D := \{\pm e_1, \pm e_2, \pm e_3\}$ and write a walk through its steps $d_j := \omega_j - \omega_{j-1} \in D$. Fix $m \geq 1$. For $j \geq m+1$ the *state* before step j is $s = (s_1, \dots, s_m) := (d_{j-1}, \dots, d_{j-m}) \in D^m$, and the *window count* is

$$N_j^{(m)} := 1 + \#\{1 \leq i \leq m : \ \omega_{j-1-i} = \omega_j\} = N(s, d_j),$$

where $N(s,d) := 1 + \#\{1 \leq i \leq m : \ -(s_1 + \cdots + s_i) = d\}$ (indeed $\omega_{j-1-i} - \omega_{j-1} = -(s_1 + \cdots + s_i)$), so that $N_j^{(m)} \leq N_j$. Let $\bar{r} : \mathbb{N}_{\geq 1} \to (0, \infty)$ be non-increasing with $\bar{r}(N) \geq \sup_{N' \geq N} \widetilde{r}(N')$, and let T_m be the operator on functions on D^m

$$(T_m f)(s) := \sum_{d \in D} \bar{r}\big(N(s,d)\big)\, f(d, s_1, \dots, s_{m-1}).$$

Lemma 5.2 (window)**.** *If $H : D^m \to (0, \infty)$ satisfies $T_m H \leq \rho' H$ pointwise, then for every $n \geq m$ and every starting point,*

$$\sum_{|\omega| = n} \prod_{j=1}^{n} \widetilde{r}(N_j) \ \leq \ (6\,\bar{r}(1))^m \, \frac{H_{\max}}{H_{\min}} \, {\rho'}^{n-m},$$

the sum running over all walks of length n in $\mathbb{Z}^3$ from that point.

Proof. For $j \leq m$ use $\widetilde{r}(N_j) \leq \bar{r}(1)$; the 6^m choices of $(d_1, \dots, d_m)$ give $(6\bar{r}(1))^m$ and run over all states. For $j > m$, $\widetilde{r}(N_j) \leq \sup_{N' \geq N_j^{(m)}} \widetilde{r}(N') \leq \bar{r}(N(s, d_j))$. Summing the last $n - m$ steps gives $(T_m^{n-m}\mathbf{1})(s) \leq (T_m^{n-m} H)(s)/H_{\min} \leq {\rho'}^{n-m} H_{\max}/H_{\min}$, since T_m has non-negative coefficients. □

The hyperoctahedral group $\mathfrak{G}$ (the 48 signed permutations of the axes) acts on D and diagonally on D^m; since $N(gs, gd) = N(s,d)$ and $(d, s_1, \dots, s_{m-1})$ is equivariant, T_m commutes with $\mathfrak{G}$. For a $\mathfrak{G}$-invariant H the inequality $T_m H \leq \rho' H$ holds everywhere as soon as it holds at one state of each orbit.

Lemma 5.3 (torus)**.** *Let $L \geq 3$. The walks of length n in $X = (\mathbb{Z}/L\mathbb{Z})^3$ from a site are in bijection with D^n, i.e. with the walks of length n in $\mathbb{Z}^3$ from a lift of that site, and the visit counts on the torus, global and in windows, are at least those of the lift. Hence the bound of Lemma 5.2 holds for walks on the torus. A walk from y to $x = y - Re_1$ in X has length at least R when $R \leq L/2$.*

Proof. Coincidences in $\mathbb{Z}^3$ project to coincidences in X. For $j > m$, $\widetilde{r}(N_j^X) \leq \bar{r}(N_j^{X,(m)}) \leq \bar{r}(N_j^{\mathbb{Z}^3,(m)})$ because $\bar{r}$ is non-increasing. The torus distance of x and y is $\min(R, L-R) = R$. □

5.3 The certificate

Proposition 5.4 (certificate, $m = 11$)**.** *Let $J_* = 40674/100000$ and let $p_1, \dots, p_{13}$ be the integers of the weights file of §9 ($p_1 = 487\,896\,913\,543$, …, $p_{13} = 78\,536\,544\,842$); put $\bar{r}_*(N) := p_{\min(N,13)}/2^{40}$. Then:*

1. *$\bar{r}_*$ is non-increasing and $\bar{r}_*(N) \geq \sup_{N' \geq N} \widetilde{r}(N'; J_*)$ for every $N \geq 1$ [certified];*
2. *there is a $\mathfrak{G}$-invariant $H : D^{11} \to \mathbb{N}$ with $H_{\min} = 663\,930\,282\,413$ and $H_{\max} = 2^{40}$ such that, for every $s \in D^{11}$,*

$$2^{30} \sum_{d \in D} p_{\min(N(s,d),13)}\, H(d, s_1, \dots, s_{10}) \ \leq \ 2639854083 \cdot 2^{40} \cdot H(s) \tag{5.1}$$

   [certified], i.e. $T_{11} H \leq \rho'_* H$ for the weights $\bar{r}_*$;

3. *$\theta_* = J_* \rho'_* = 53686712485971/53687091200000 < 1$ [certified].*

The computations and their verification are described in §6.

5.4 Proof of Theorem 2.1

Let $\beta \in [0, \beta_*]$, so $J = \beta/2 \le J_* \le 0.44$; for $J = 0$ the loop expectation vanishes, so let $J > 0$. By (4.3) it suffices to bound $\widetilde{G}(x,y) = \sum_n J^n S_n$, $S_n := \sum_{\omega: y \to x,\, |\omega| = n} \prod_{j=0}^{n} \widetilde{r}(N_j; J)$. By Lemma 5.1, $J\widetilde{r}(N;J) \le J_* \widetilde{r}(N;J_*)$ for all N, so for every walk $J^n \prod_{j=0}^{n} \widetilde{r}(N_j;J) \le \widetilde{A}_1(J) \prod_{j=1}^{n} J_* \widetilde{r}(N_j;J_*)$; only the exact weights are compared, never the rounded ones. By Lemmas 5.2 and 5.3 with $\widetilde{r}(\cdot;J_*)$, $\bar{r}_*$, H and ρ'_* (Proposition 5.4), for $n \ge m = 11$,

$$J^n S_n \le \widetilde{A}_1(J)\, (6 J_* \bar{r}_*(1))^m \, \frac{H_{\max}}{H_{\min}} \, (J_* \rho'_*)^{n-m} = \widetilde{A}_1(J)\, K_0\, \theta_*^n,$$

$$K_0 := \Big(\frac{6\, \bar{r}_*(1)}{\rho'_*} \Big)^{11} \frac{H_{\max}}{H_{\min}} = 3.978058961\ldots. \tag{5.2}$$

For $n < m$, bounding each step by $6 J_* \bar{r}_*(1)$ gives $J^n S_n \le \widetilde{A}_1(J) (6\bar{r}_*(1)/\rho'_*)^n \theta_*^n \le \widetilde{A}_1(J) K_0 \theta_*^n$, because $6\bar{r}_*(1)/\rho'_* = 1.0829\ldots > 1$ and $H_{\max}/H_{\min} \ge 1$. Walks from y to x have length $n \ge R$ (Lemma 5.3) and $\widetilde{A}_1(J) \le a_1 = \frac{1}{2}$, so $\widetilde{G}(x,y) \le \frac{1}{2} K_0\, \theta_*^R/(1-\theta_*)$, and (4.3) gives $|\langle W_\gamma \rangle| \le (K_0/(1-\theta_*))^T \theta_*^{RT}$. □

Remark 5.5 (the two improvements separately, and the role of m). With the weights $r(N)$ of Borgs and Seiler, the same operators give the spectral radii $\rho_1 = 17/6$ (the bound of §3.5), $\rho_3 = 2.7869954984$, $\rho_5 = 2.7653748962$, $\rho_7 = 2.7528722513$, $\rho_9 = 2.7447464071$, $\rho_{11} = 2.7390488353$; (II) alone therefore gives $\beta < 2/\rho_{11} = 0.73018$ [numerical]. With the weights of (I), Table 3 lists the thresholds; the increments decrease slowly with m [numerical].

m	orbits	J_*	β_*	ρ'_*	status
1	1		0.7824		numerical
3	11		0.7972		numerical
5	236		0.8044		numerical
7	6 896	$40432/10^5$	0.80864	$2655621200/2^{30}$	certified
9	226 496	$40574/10^5$	0.81148	$2646370548/2^{30}$	certified
11	7 821 056	$40674/10^5$	0.81348	$2639854083/2^{30}$	certified (Theorem 2.1)
13	276 294 656	$40749/10^5$	0.81498	$2635005744/2^{30}$	certified (Remark 2.3)

Table 3: Thresholds of the combined method as a function of the memory m. For each certified row, $J_* + 10^{-5}$ fails the test $J\rho' < 1$ of the dichotomy (grid 10^{-5} in J). The numbers of orbits are the Burnside counts $(6^m + 3 \cdot 4^m + 15 \cdot 2^m)/48$.

5.5 A tilted bound (new in version 1.1)

The proof of Theorem 2.1 bounds $\widetilde{G}(x,y)$ by a sum over *all* walks of length $n \ge R$ from y, whatever their end point. Near the threshold this costs the factor $1/(1-\theta_*)$, and the rate obtained, $-\log \theta_*$, is of order $1 - \theta_*$. Weighting each step by $e^{\eta \langle \hat{e}, d \rangle}$, where $\hat{e}$ is the direction from y to x (as for the tilted walk in the proof of [16, Thm. 7.1]), keeps the threshold and gives a rate of order $\sqrt{1-\theta_*}$ [numerical], as Table 4 shows; the parity of walks in $\mathbb{Z}^3$ halves the constant on even tori. The weights, the window counts and Lemmas 5.1, 5.2 and 5.3 are those of §5; only the transfer operator and the certificate change. We write η for the tilt and σ_η for the rate, to avoid confusion with the parameters λ_z of the walk expansion and with the string tension $\sigma(\beta)$.

For $\hat{e} \in D$ and $\eta \ge 0$, let $T_m^{(\eta,\hat{e})}$ act on functions on D^m by

$$(T_m^{(\eta,\hat{e})} f)(s) := \sum_{d \in D} \bar{r}(N(s,d))\, e^{\eta \langle \hat{e}, d \rangle}\, f(d, s_1, \ldots, s_{m-1}).$$

Lemma 5.2 holds for $T_m^{(\eta,\hat e)}$ with the same proof: if $T_m^{(\eta,\hat e)}H \le \rho' H$, the sum over the walks of length $n \ge m$ from a point of $\prod_{j=1}^n \widetilde r(N_j)\, e^{\eta\langle \hat e, d_j\rangle}$ is at most $\big(\bar r(1)(e^\eta + e^{-\eta} + 4)\big)^m (H_{\max}/H_{\min})\, \rho'^{\,n-m}$, because $\sum_{d\in D} e^{\eta\langle \hat e, d\rangle} = e^\eta + e^{-\eta} + 4$; and Lemma 5.3 transfers this bound to tori. For $g \in \mathfrak G$ one has $N(gs, gd) = N(s,d)$ (N only involves the equalities $-(s_1 + \cdots + s_i) = d$, which the linear map g preserves) and $\langle g\hat e, gd\rangle = \langle \hat e, d\rangle$; so $T_m^{(\eta,g\hat e)}$ is conjugate to $T_m^{(\eta,\hat e)}$ by $s \mapsto gs$, and if $T_m^{(\eta,e_1)}H \le \rho' H$ then $T_m^{(\eta,ge_1)}(H\circ g^{-1}) \le \rho'\, H\circ g^{-1}$, with the same ratio $H_{\max}/H_{\min}$.

Theorem 5.6 (tilted bound)**.** *Let $m \ge 1$ and $0 < J_i \le 0.44$; let $\bar r$ be non-increasing with $\bar r(N) \ge \sup_{N'\ge N} \widetilde r(N'; J_i)$ for all $N \ge 1$; let $\eta \ge 0$; and let $H : D^m \to (0,\infty)$ and $\rho' > 0$ satisfy $T_m^{(\eta,e_1)}H \le \rho' H$. Put*

$$\theta := J_i \rho', \qquad \sigma_\eta := \eta - \log\theta, \qquad q := \frac{\bar r(1)\,(e^\eta + e^{-\eta} + 4)}{\rho'},$$

$$K_0^{(\eta)} := \max(1,q)^m \frac{H_{\max}}{H_{\min}}, \qquad C^{(\eta)} := \frac{K_0^{(\eta)}}{1-\theta^2}, \qquad C_{\mathrm{odd}}^{(\eta)} := \frac{K_0^{(\eta)}}{1-\theta},$$

and assume $\theta < 1$. Let $0 \le \beta \le 2J_i$, and let γ be a rectangular loop with sides R along e_μ and T along e_ν.

1. *On the torus $(\mathbb Z/L\mathbb Z)^4$ with L even and $L \ge \max(4, 2R, T+1)$:*

$$|\langle W_\gamma\rangle_{L,\beta}| \;\le\; \big[C^{(\eta)}(1 + e^{-\sigma_\eta(L-2R)})\big]^T e^{-\sigma_\eta RT} \;\le\; \big(2C^{(\eta)}\big)^T e^{-\sigma_\eta RT}.$$

2. *On the torus with L odd and $L \ge \max(3, 2R+1, T+1)$: the same bound with $C_{\mathrm{odd}}^{(\eta)}$ in place of $C^{(\eta)}$.*

3. *Every weak limit point of $(\mu_{L,\beta})$ along even $L \to \infty$ satisfies $|\langle W_\gamma\rangle| \le \big(C^{(\eta)}\big)^T e^{-\sigma_\eta RT}$.*

Status: [proved], given H and ρ'; the instances of Table 4 are [certified] (Proposition 5.7). Not formalised in Lean.

Proof. Let $J = \beta/2 \in (0, J_i]$ (for $J = 0$ the expectation vanishes), and let $\hat e \in D$ be the direction from y to x in (4.3), so that $x = y + R\hat e$. As in §5.4, Lemma 5.1 applied to the exact weights gives, for every walk of length n, $J^n \prod_{j=0}^n \widetilde r(N_j; J) \le \widetilde A_1(J) \prod_{j=1}^n J_i\, \widetilde r(N_j; J_i)$.

Walks in $\mathbb Z^3$. A walk from y to $y + R\hat e$ satisfies $\prod_{j=1}^n e^{\eta\langle \hat e, d_j\rangle} = e^{\eta R}$, $n \ge R$ and $n \equiv R \pmod 2$, since $\mathbb Z^3$ is bipartite. Multiplying each term by $e^{-\eta R}\prod_j e^{\eta\langle \hat e, d_j\rangle} = 1$ and dropping the constraint on the end point (all terms are non-negative),

$$\sum_{\omega: y \to y + R\hat e} J^{|\omega|} \prod_{j=0}^{|\omega|} \widetilde r(N_j; J) \;\le\; \widetilde A_1(J)\, e^{-\eta R} \sum_{\substack{n \ge R \\ n \equiv R\,(2)}} J_i^n \sum_{|\omega|=n} \prod_{j=1}^{n} \widetilde r(N_j; J_i)\, e^{\eta\langle \hat e, d_j\rangle}, \tag{5.3}$$

the inner sum running over all walks of length n from y. Write $\hat e = ge_1$ with $g \in \mathfrak G$. By the tilted form of Lemma 5.2 with the vector $H\circ g^{-1}$, $J_i^n \sum_{|\omega|=n} \cdots \le J_i^n (q\rho')^m (H_{\max}/H_{\min})\rho'^{\,n-m} = q^m (H_{\max}/H_{\min})\,\theta^n \le K_0^{(\eta)}\theta^n$ for $n \ge m$. For $n < m$, bounding every step by $\bar r(1)(e^\eta + e^{-\eta} + 4) = q\rho'$ gives $q^n\theta^n \le \max(1,q)^m\theta^n \le K_0^{(\eta)}\theta^n$. Summing the geometric series over $n \ge R$, $n \equiv R$, and using $\widetilde A_1 \le \tfrac12$, the right side of (5.3) is at most $\tfrac12 K_0^{(\eta)} e^{-\eta R}\theta^R/(1-\theta^2) = \tfrac12 C^{(\eta)} e^{-\sigma_\eta R}$.

Tori. By Lemma 5.3, a walk from y to x in $X = (\mathbb Z/L\mathbb Z)^3$ lifts to a unique walk in $\mathbb Z^3$ from a fixed lift of y, ending at a lift of x, i.e. with displacement $R\hat e + Lk$, $k \in \mathbb Z^3$, and the window bound applies with the visit counts of the lift (the counts on the torus are larger and $\bar r$ is non-increasing). If $\langle \hat e, k\rangle \ge 0$, the displacement along $\hat e$ is at least R, so

$e^{-\eta R}\prod_j e^{\eta\langle\hat e,d_j\rangle}\ge 1$ and these walks are bounded as in (5.3). If $\langle\hat e,k\rangle\le -1$, the displacement along $\hat e$ is $R+L\langle\hat e,k\rangle\le R-L\le -R$, so the length is at least $L-R$ and $e^{-\eta(L-R)}\prod_j e^{\eta\langle-\hat e,d_j\rangle}=e^{-\eta(L-R)-\eta(R+L\langle\hat e,k\rangle)}\ge 1$; inserting this factor and using $T_m^{(\eta,-\hat e)}$, the same argument bounds these walks by $\frac12 C^{(\eta)}e^{-\sigma_\eta(L-R)}$. For even L the coordinates of Lk are even, so $n\equiv R \pmod 2$ for every lift and $L-R\equiv R$. Therefore $\widetilde G(x,y)\le \frac12 C^{(\eta)}e^{-\sigma_\eta R}(1+e^{-\sigma_\eta(L-2R)})$, and (4.3) gives item 1. For odd L the parity is not used, $\sum_{n\ge N}\theta^n=\theta^N/(1-\theta)$, and the same argument gives item 2. Item 3 follows from item 1, since $e^{-\sigma_\eta(L-2R)}\to 0$ and W_γ is a bounded continuous function of finitely many links. □

Proposition 5.7 (tilted certificates, $m=11$)**.** *For each row of Table 4, with $J_i=\beta_i/2$, the weights $\bar r(N)=p_{\min(N,13)}/2^{40}$ of the weights file at J_i (for J_* that of Proposition 5.4, for the other rows those of Table 2), $e^\eta=a/b$ and the stated integer P, there is $H:D^{11}\to\mathbb{N}$ with $H_{\max}=2^{40}$ and the stated $H_{\min}$ such that, for every $s\in D^{11}$,*

$$\sum_{d\in D} c_d\, p_{\min(N(s,d),13)}\, H(d,s_1,\dots,s_{10}) \;\le\; P\cdot 2^{10}\, ab\, H(s), \tag{5.4}$$

where $c_{+e_1}=a^2$, $c_{-e_1}=b^2$ and $c_d=ab$ for the four other steps d; that is, $T_{11}^{(\eta,e_1)}H\le (P/2^{30})H$ [certified]; moreover $\theta=J_iP/2^{30}<1$ [certified]. Consequently Theorem 5.6 holds with $\rho'=P/2^{30}$ and the constants of Table 4.

$\beta\le$	e^η	P	$H_{\min}$	$1-\theta$	$\sigma_\eta\ge$	$C^{(\eta)}\le$	$R^{\lim}_{\min}\le$	$R^{\mathrm{tor}}_{\min}\le$
0.81348	201/200	2 639 868 799	663 255 845 757	$1.4795\cdot10^{-6}$	0.004989	$1.3457\cdot10^{6}$	2 829	3 107
0.81348	251/250	2 639 863 512	663 389 751 364	$3.4823\cdot10^{-6}$	0.003995	$5.7164\cdot10^{5}$	3 318	3 665
0.81	10/9	2 648 763 350	647 937 564 146	$9.2449\cdot10^{-4}$	0.1062	2 192.6	72.4	85.5
0.80	5/4	2 678 382 511	632 089 802 916	$2.2247\cdot10^{-3}$	0.2253	915.6	30.3	36.5
0.70588	7/4	2 898 427 727	568 404 757 891	$4.7283\cdot10^{-2}$	0.6080	43.8	6.3	8.5

Table 4: Tilted certificates with memory 11 (full state space D^{11}, no symmetry reduction) and the constants of Theorem 5.6. The exact θ is the rational $J_iP/2^{30}$; $1-\theta$ and σ_η are rounded down; $C^{(\eta)}$, $R^{\lim}_{\min}:=\log C^{(\eta)}/\sigma_\eta$ (weak limits) and $R^{\mathrm{tor}}_{\min}:=\log(2C^{(\eta)}_{\mathrm{odd}})/\sigma_\eta$ (every torus, even or odd) are rounded up; beyond these radii the bounds are below 1. For comparison, Theorem 2.1 and Table 2 give $\sigma\ge 7.05\cdot10^{-6}$, 0.0034, 0.0133, 0.1159 and $R_{\min}\approx 1.9\cdot10^{6}$, $2.1\cdot10^{3}$, 430, 31 (every torus) for $\beta\le 0.81348$, 0.81, 0.80, 0.70588.

Corollary 5.8 (string tension, version 1.1)**.** *With $\sigma(\beta)$ as in Corollary 2.2: $\sigma(\beta)\ge 0.004989$ for $0<\beta\le\beta_*$, $\sigma(\beta)\ge 0.1062$ for $0<\beta\le 0.81$, $\sigma(\beta)\ge 0.2253$ for $0<\beta\le 0.80$, and $\sigma(\beta)\ge 0.6080$ for $0<\beta\le 0.70588$.*

Proof. Items 1 and 2 of Theorem 5.6 cover every $L\ge\max(3,2R,T+1)$ (for odd L, $L\ge 2R$ is the same as $L\ge 2R+1$), with the constant $C':=2C^{(\eta)}_{\mathrm{odd}}\ge 2C^{(\eta)}$, which depends neither on L nor on T. Hence $-\frac{1}{RT}\log|\langle W_{R\times T}\rangle_{L,\beta}|\ge\sigma_\eta-(\log C')/R$; let $R\to\infty$ and use Table 4. □

Remark 5.9 (scope). Like Theorem 2.1, Theorem 5.6 is proved on tori and for weak limits of torus measures. Free spatial boundaries and a non-periodic cutting direction are treated neither for Theorem 2.1 nor for Theorem 5.6 (§8.1); for the tilted operator, the compensation at a free boundary that such an extension would need has not been worked out. The weights files are those of version 1.0 (Proposition 5.4 and Table 2), used without recomputation; `c1_poids_exacts.py` regenerates them byte for byte.

6 The computer-assisted part

6.1 Exact enclosures of the weights

Put $u := 1-\lambda$, so that $M_J(u^2) = \sum_k c_k J^{2k} u^{2k}$, $c_k = 1/(k!(k+1)!)$. Truncate at $K = 10$: for $s \in [0,1]$, $0 \le M_J(s) - M_{J,K}(s) \le \tau := c_{11} J^{22}/(1 - J^2/156)$, because $c_{k+1}J^2/c_k = J^2/((k+1)(k+2)) \le J^2/156$ for $k \ge 11$. Then

$$\Big(\frac{M_{J,K}(u^2)}{M_{J,K}(1)+\tau}\Big)^3 \le h_J(\lambda) \le \Big(\frac{M_{J,K}(u^2)+\tau}{M_{J,K}(1)}\Big)^3.$$

The cubes are polynomials in u with rational coefficients, and $\int_0^1 (1-u)^{n-1} u \cdot u^j \, du/(n-1)! = (j+1)!/(n+j+1)!$ exactly. This gives rationals $\widetilde{A}_n^{\inf} \le \widetilde{A}_n \le \widetilde{A}_n^{\sup}$ for $n \le 12$ (relative width below 10^{-24} at J_*). Then $u_1 := \widetilde{A}_1^{\sup}$, $u_N := \min(\widetilde{A}_N^{\sup}/\widetilde{A}_{N-1}^{\inf}, 1/(N+1))$ for $2 \le N \le 12$, $u_{13} := 1/14$ (which bounds $\widetilde{r}(N)$ for all $N \ge 13$ by Lemma 4.6), $\bar{W}_n := \max(u_n, \dots, u_{13})$ and $p_n := \lceil 2^{40} \bar{W}_n \rceil$. By construction $\bar{r}_*(N) = p_{\min(N,13)}/2^{40} \ge \sup_{N' \ge N} \widetilde{r}(N')$, which is item 1 of Proposition 5.4. Everything is computed with Python fractions (program `c1_poids_exacts.py`).

6.2 The integer Perron certificate

The program `tm_cert.c` enumerates one representative per orbit of $\mathfrak{G}$ on D^m (canonical form, with s_1 the most recent step: $s_1 = +e_1$; the first s_i off the e_1-axis equal to $+e_2$; the first s_i along the third axis equal to $+e_3$), computes the transitions and window counts, runs a floating-point power iteration, and rounds the approximate Perron vector to integers $H := \max(1, \lfloor 2^{40} h/h_{\max} \rfloor)$. It then sets $A := \max_s \lceil 2^{30} \sum_d p_{\min(N(s,d),13)} H(s'_d)/(2^{40} H(s)) \rceil$ (exact integer division) and re-checks (5.1) for every representative in unsigned 128-bit integer arithmetic; $\rho' = A/2^{30}$. Floating point only serves to find H; the inequality is checked exactly. For $m = 11$ there are 7 821 056 representatives and $A = 2639854083$.

Exhaustiveness of the representatives. The table is closed under transitions (every successor state lies in a represented orbit), and from any state every state is reached in exactly m transitions (choose the m new steps); so a non-empty closed table meets every orbit. In addition, its size equals the Burnside count $(6^m + 3 \cdot 4^m + 15 \cdot 2^m)/48$ and no two entries lie in the same orbit.

6.3 Independent verifications

- *Second program.* `verif_cert.c`, written by the same author as `tm_cert.c` but with independent implementation choices, re-verifies the certificate: an orbit is identified by the minimal base-6 code among its 48 images under explicitly constructed group elements (not by the canonical form of `tm_cert`); window counts are recomputed by explicit tracking of positions; it checks that the entries lie in distinct orbits, that every transition lands in a represented orbit, that $H \ge 1$, and (5.1) in 128-bit integers. Result for $m = 11$: zero violations, zero missing transitions (also for $m = 7, 9, 13$).

- *Full state space, $m = 11$.* A third program, written in a separate session for an adversarial verification without reusing `tm_cert` or `verif_cert`, extends H to all $6^{11} = 362\,797\,056$ raw states through orbit look-up and checks (5.1) state by state, with no symmetry argument: zero violations, largest ratio of the two sides 0.999999999788. It also recomputes the Perron root on the raw states: $2.4585557007\ldots < \rho'_*$ [numerical].

- *Memory* 13. `verif_cert` (zero violations), the third program (all 276 294 656 representatives and $2 \cdot 10^8$ random raw states), and a fourth program written by another reviewer (all representatives, 10^7 random raw states, Burnside count, weights recomputed in exact rationals with bounded tails): no violation; largest ratio 0.9999999999166.

- *Weights.* The bounds $\bar{r}_*(N) \geq \sup_{N' \geq N} \widetilde{r}(N'; J_*)$ were re-derived by a different method (ball arithmetic in Arb [12]: series of the cube in $(1-\lambda)^2$ with exactly known total sum, $M_J(1)$ from Arb's Bessel functions, and a rigorous comparison with $p_n/2^{40}$) for $m = 11$, $m = 13$ and all rows of Table 2; and independent quadratures (tanh–sinh and Gauss–Legendre) reproduce p_1, p_2, p_3 after rounding [numerical].
- *Arithmetic of item 3.* $J_*\rho'_* < 1$ is checked with exact fractions.

6.4 A guard against silent truncation

In python-flint, operations on power series (`arb_series`) are silently truncated at the length `ctx.cap`, whose default is 10, below the 61 coefficients of the cube of a polynomial of degree 20. The cross-check `c2_poids_flint.py` computes the enclosures of §6.1 in Arb both with polynomials (no truncation) and with series for `cap` $\in \{10, 30, 60, 61, 80\}$, and requires that the large caps reproduce the polynomial computation and that `cap` $= 10$ differs from it: the relative deviation is $3.8 \cdot 10^{-7}$ at `cap` $= 10$ (truncation detected) and 0 at `cap` ≥ 60. The Arb balls (200 bits, relative radius below $8 \cdot 10^{-59}$) have centres within 10^{-14} of the exact rationals of §6.1; this is a cross-check, the certification being the exact rational computation. The independent Arb computation of §6.3 fixes `ctx.cap` explicitly and detects the same truncation (a series of length 10 instead of 40).

6.5 Validation of the counting programs

With the weights of Borgs and Seiler, $r(1) = \frac{1}{2}$, $r(N) = 1/(N+1)$, the programs reproduce $\rho_1 = 17/6$, hence $12/17$, and the values $\rho_3, \ldots, \rho_{11}$ of Remark 5.5; these values coincide with those of a program written independently (see the statement on the use of AI), and the numbers of orbits $1, 3, 11, 48, 236, 1248, 6896, 39168, 226496, 1325568, 7821056$ ($m = 1, \ldots, 11$) are the Burnside counts.

6.6 Tilted certificates (version 1.1)

The program `cert_incline11.c` works on the full state space D^m ($6^{11} = 362\,797\,056$ states for $m = 11$, no symmetry reduction; base-6 code with the most recent step s_1 as leading digit). It computes the window counts, runs a floating-point power iteration for $T_m^{(\eta,e_1)}$ (50 to 57 iterations for the runs of Table 4), rounds the vector up, $H := \max(1, \lceil 2^{40} h/h_{\max} \rceil)$, sets $P := \lceil 2^{30} \rho_{\mathrm{fl}}(1 + 10^{-9}) \rceil + 1$, where ρ_{fl} is the largest ratio $(T_m h)(s)/h(s)$ of the last iteration, and checks (5.4) for every state in unsigned 128-bit integers (both sides are below 2^{107} for $a, b \leq 4096$). For $m = 11$ a run takes 34 to 39 s on 8 cores and about 11 GB of memory. With the weights of Borgs and Seiler and $\eta = 0$ it reproduces $\rho_1 = 17/6$, ρ_3 and ρ_5 of Remark 5.5.

A second program, `b7_verif11.py`, re-checks each certificate on all states. It was written separately by the same author as `cert_incline11.c`, but with independent implementation choices (as for `verif_cert` in §6.3): the states are indexed with the opposite digit order and H is read through the digit permutation; successors and window counts are recomputed from the digits, the counts by explicit tracking of positions; and (5.4) is checked exactly with 22-bit limbs in 64-bit integers, without 128-bit arithmetic. For the five certificates of Table 4 it finds zero violations on all $362\,797\,056$ states; the smallest relative margins correspond to 4.05 to 4.56 units of P (floating-point estimate), and with $P - 5$ it reports violations for each of them; with P lowered by 10^{-5} in relative terms it reports violations as well. The numbers θ, $K_0^{(\eta)}$, $C^{(\eta)}$ and $C_{\mathrm{odd}}^{(\eta)}$ are computed in exact rationals, and σ_η and the logarithms in Arb balls at 128 and 256 bits, which are required to overlap (`b7_constantes.py`); a second program written separately during the review of this version (`checks/table4_fractions.py`: exact fractions and 60-digit decimal logarithms, without Arb) gives the same values.

The six vectors H (2 902 376 448 bytes each: the five certificates of Table 4 and the run at $\eta = 0$ of Remark 6.1) are supplied, compressed with `xz`, in the Zenodo record of this version; they are too large for GitHub. `cert_incline11` also regenerates them byte for byte, in about 40 s each on 8 cores, on x86-64 with gcc and no machine-specific options (checked for all six: the regenerated vectors have the listed fingerprints); their SHA-256 are listed in `supplement_v1.1`. On another platform the floating-point iteration may round differently; a vector obtained in this way is again a valid certificate if it passes the exact check, but its $H_{\min}$, its P and the constants of Table 4 may then differ slightly.

Remark 6.1 (two certificates at $\eta = 0$). Run with $\eta = 0$ at J_*, `cert_incline11` returns $P = 2\,639\,854\,087$ and $H_{\min} = 663\,930\,282\,414$, whereas Proposition 5.4 has $A = 2\,639\,854\,083$ and $H_{\min} = 663\,930\,282\,413$. The difference comes from the choice of the integer, not from the operator or the weights. The floating-point Perron root is $\rho_{\mathrm{fl}} = 2.45855570066\ldots$, the value found on the raw states in §6.3; thus $2^{30}\rho_{\mathrm{fl}} = 2\,639\,854\,082.44$, the relative margin of 10^{-9} gives $2\,639\,854\,085.08$, and rounding up and adding 1 gives P. `tm_cert` instead takes the smallest integer admissible for its vector (§6.2). With the vector of `cert_incline11` the smallest admissible integer is also A: `b7_verif11.py` finds zero violations on all 6^{11} states with $P = A$, and violations with $P = A - 1$. The two vectors come from different floating-point iterations (on orbits in `tm_cert`, on raw states here) and are rounded in opposite directions (down in `tm_cert`, up here), hence the difference of one unit in $H_{\min}$. Both certificates are valid: the published one was re-checked for version 1.1 with `verif_cert` (zero violations, "CERTIFICAT VALIDE"; `supplement_v1.1/data/verif_cert_published_m11.log` records the command and the SHA-256 of the file checked), and the new one with `b7_verif11.py` (zero violations on all 6^{11} states). Theorem 2.1 uses the published certificate.

7 Consistency tests (not part of the proof)

7.1 One-dimensional slabs

On the chain $\mathbb{Z}$, a tree, the ratio of partition functions factorises exactly, $Z(\boldsymbol{J}^{\lambda})/Z(J) = \prod_{\ell} \rho_J(s_\ell)$, so (3.3) with the exact factor must reproduce the exact correlation $\mathbb{E}[q_0 \cdot q_R] = u^R$, $u := I_2(2J)/I_1(2J)$; these chains are the slabs of the two-dimensional theory. For two sites this identity was checked analytically (the walks have length $2k+1$ and the sum reduces to $(J/I_1(2J)) \sum_n J^{2n+1}/(n!(n+2)!) = I_2(2J)/I_1(2J)$). For $R \le 4$ and $J \in \{0.2, 0.4, 0.6\}$ the sum over *all* walks, organised by edge-crossing numbers without truncation in length, agrees with u^R to $9 \cdot 10^{-29}$, $4 \cdot 10^{-21}$ and $9 \cdot 10^{-17}$ respectively (truncation of the series in λ) [numerical]. With the Cauchy–Schwarz factor adapted to two edges per site, $h = \rho_J((1-\lambda)^2)$, the bound $2\widetilde{G}(R)$ exceeds u^R by the factors of Table 5: (I) is not too strong, and its loss is small in this case.

$J \backslash R$	1	2	3	4
0.2	1.00443	1.00554	1.00665	1.00776
0.4	1.01755	1.02196	1.02639	1.03083
0.6	1.03889	1.04871	1.05862	1.06863

Table 5: Ratio $2\widetilde{G}(R)/u^R$ on the chain $\mathbb{Z}$, with the site-by-site factor $h = \rho_J((1-\lambda)^2)$ [numerical].

7.2 The $O(4)$ model at order J^3: sensitivity of (I)

For the untwisted $O(4)$ model on $\mathbb{Z}^3$ with weight $\exp(2J \sum q_x \cdot q_y)$, the nearest-neighbour correlation is $\mathbb{E}[q_0 \cdot q_{e_1}] = u + 4u^3 + O(J^5) = \frac{J}{2} + \frac{5}{12}J^3 + O(J^5)$ (the four paths of length three

that close a plaquette with the edge each contribute u^3; checked on a ring of four sites by a character expansion). Exact symbolic computation of the walk expansion at this order gives: with the exact factor $c(\lambda)$, $\frac{J}{2} + \frac{5}{12}J^3$ (equality); with the site-by-site factor $h = \rho^3$, $\frac{J}{2} + \frac{23}{36}J^3$, above the true value; with $h = \rho^k$ the coefficient is $\frac{25}{18} - \frac{k}{4}$, below the true value for $k > 35/9$. An error of one whole edge per site in Corollary 4.5 (e.g. ρ^4) would therefore be detected at this order; an error of half an edge would not.

7.3 The limit of infinite magnetic coupling

In the finite-temperature setting of [2] (electric and magnetic couplings J_E, J_M, L_0 time layers), the inequalities of §§4–5 involve only the electric coupling and the positivity of the transfer matrix. For $L_0 = 1$ and $J_M \to \infty$ the horizontal links become pure gauge and a slab becomes the untwisted $O(4)$ model on $\mathbb{Z}^3$ with coupling $2J_E$ per edge ($2J = \beta$ in the isotropic case); the argument then asserts exponential decay of its two-point function for $2J_E < 0.81348$ (or 0.81498). This is consistent with the Monte Carlo estimates of the critical coupling of the $O(4)$ model on the simple cubic lattice with action $-\beta \sum s_x \cdot s_y$, $\beta_c = 0.935861(8)$ (the Monte Carlo value retained in [3, Tables 3–4]) and 0.9360(1) [13]: the margin is a factor 1.15. This case is not a statement of this note; it is a sanity check, since no bound that is uniform in J_M can hold beyond the critical coupling of the $O(4)$ model.

8 Discussion

8.1 Scope and limitations

- *Tori only.* Corollary 4.5 uses that every site of the spatial torus has exactly six edges, and the trace formula (3.5) uses periodicity in the cutting direction. For other boundary conditions the factor h must be adapted (sites with $k < 6$ edges receive $\rho^{k/2} \geq \rho^3$), which we have not done. In infinite volume we only obtain the weak limit points of torus measures; by contrast the range $\beta < 2/3$ of [16, Thm. 7.1] holds for every Gibbs measure.

- *Loops and constants.* Only planar rectangular loops are treated, with $R \leq L/2$ and $T \leq L - 1$. The string tension of Theorem 2.1 is tiny near the threshold ($7 \cdot 10^{-6}$ at β_*), and the bound is non-trivial only for $R > R_{\min} \approx 1.9 \cdot 10^6$ there. Theorem 5.6 (version 1.1) raises the lower bound on the tension at β_* to 0.004989 and lowers $R_{\min}$ to about $3.1 \cdot 10^3$ on every torus ($2.8 \cdot 10^3$ in the weak limits); the method still gives a range rather than a good tension (Remark 2.4).

- *Tilted bound.* Theorem 5.6 is proved, like Theorem 2.1, on tori (even and odd) and for weak limits; free spatial boundaries and a non-periodic cutting direction are not treated (Remark 5.9). It is not formalised in Lean; its certificates (2.9 GB each) are supplied in compressed form in the Zenodo record and can be regenerated (§6.6).

- *Verification.* The proof is computer-assisted and has been checked by machine only (§1.4). The certificates for $m = 11$ (Proposition 5.4 and the tilted ones) are supplied; the one for $m = 13$ is described by its hash.

- *Limits of the method* [numerical]. With the weights of Borgs and Seiler, an exact enumeration of the walks up to length 14 and an extrapolation give a growth rate $\mu_\infty \approx 2.726$–2.731, so that the count alone is limited to about 0.733; with the weights of (I), the certified thresholds of Table 3 appear to converge slowly to about 0.82. Memory 15 would require about 10^{10} orbits. Replacing the worst case of six visited neighbours in Corollary 4.5 by the actual number would give 0.8060 instead of 0.7824 at memory 1;

this information is not local along the walk, and we have not found a finite-memory rule that uses it.

- *Not used.* Borgs and Seiler remark that the perimeter decay of the characters $\operatorname{Tr} U_{C_m}$ could be used to sharpen their bound on J [2, second (B.14), p. 377]; they give no number, and we do not use it. The improvements only involve the electric coupling and the positivity of the transfer matrix, so they should apply to the finite-temperature setting of [2] as well; we have not written this out.

8.2 Comparison with heuristic values

Fröhlich's lecture notes state, "in mean" [9, p. 24, (6.14)–(6.16)] or "in measure" [10, p. 20, (34)–(35)], a decay estimate $F \lesssim e^{-\alpha|\gamma|}$ for the slab factors of a walk expansion, with $\alpha > -\ln[(2D-3)(D-1)(4D-4)^{-1}\beta_0]$, and conclude area decay for $\beta_0 < 4/5$ when $D = 4$ [read]. [9, p. 24] adds that "(6.14) is only a rough estimate"; [10, p. 20] attributes the estimate to the preprint of [6], whose refined estimates are stated with proofs announced to "appear elsewhere" [6, preprint pp. 55, 57]. We know of no proof, and treat 4/5 as a heuristic value.

For $D = 4$ the condition reads $5 \cdot \beta_0/4 < 1$. Here $5 = 2D - 3$ is the number of non-backtracking continuations of a walk in $\mathbb{Z}^3$, and $\beta_0/4$ is the weight of a step to a new site in the walk expansion when β_0 is the coupling of the slab σ-model. In the true slab model this coupling is β (Lemma 3.1), as follows after correcting the factor 2 of [6] discussed in [16, §7.2]; in this reading the announced range is $\beta < 4/5$. Read literally, with β_0 the coefficient of $\operatorname{Re}\operatorname{Tr}$ ([10, p. 22] writes $\beta \equiv 2\beta_0$), it would be $\beta < 8/5$.

In the first reading, $5\beta/4 < 1$ is the small-β linearisation of the following *non-backtracking heuristic*, which we formulate here (we do not attribute it to a source)

$$5\,u(\beta) < 1, \qquad u(\beta) := \frac{I_2(\beta)}{I_1(\beta)} = \frac{\beta}{4} - \frac{\beta^3}{96} + O(\beta^5),$$

where $u(\beta)$ is the exact correlation $\mathbb{E}[q_x \cdot q_y]$ across a single edge of coupling β; $5u(\beta) = 1$ at $\beta = 0.822225$ [numerical]. Our rigorous range 0.81348 (0.81498 with memory 13) lies above 4/5 and below 0.8222 (at β_*, $5u(\beta_*) = 0.9899$), and the thresholds of Table 3 seem to approach about 0.82. The new number is therefore best read as a rigorous realisation of most of a known heuristic, rather than as an unexpected value.

9 Reproducibility

The supplementary folder contains the programs and data below; comments and messages in the programs are in French. SHA-256 fingerprints:

`certification/c1_poids_exacts.py`
`8efd0508f3e21ef0db82df056cfc96b6154d8d14a255590feb4dab7a0b618396`

`certification/c2_poids_flint.py`
`6c82f6c30d24a2d993be06fbce6f219f5af4587a80756b94a9db6c1a33c3a939`

`certification/c3_seuil.py`
`6c1d5397fa2a5f4dd43b4f0f01437c65d8cf947d22871c988b2dc2af27a0fab1`

`certification/tm_cert.c`
`acf257f9344311e74f04540aff7f95cefb55fc8a8e34296e2080cb12fcc98749`

`certification/verif_cert.c`
`ebdfacb544f967480c8849da6fad8603f4afa1018dd740793a74a4f2bade33ca`

`data/poids_m11_J40674_100000.txt`
`28c789d9a8096a71bc46ba8a9a8bde25e360af4896e0e8f4c6964e9448419317`

`data/certificat_h_m11_J40674_100000.bin.xz` (8 780 360 bytes, `xz -9e`)
`776f4f27be0aabb32ec81050883448b4004db11289b2a1cf338614e188bbc1e4`

`data/certificat_h_m11_J40674_100000.bin` (125 136 920 bytes, unpacked)
`b50d3769b0ea17d1da13f4e870a7786407192a9885ecceb390ef18cde50cf142`

`data/poids_m13_J40749_100000.txt`
`fcdc6bc36d8fd9de3da60bee15c421ca0a2b0e4119f403dbd39d21085b63c706`

`certificat_h_m13_J40749_100000.bin` (4 420 714 520 bytes, not supplied)
`c8ad0deb4a5ec86f8fb3cc5802feaee9a3e1273cb8e14883ccf52ff4590082c7`

*The memory-*11 *certificate* is supplied in compressed form, as `data/certificat_h_m11_J40674_100000.bin.xz`; running `xz -dk certificat_h_m11_J40674_100000.bin.xz` in `data/` recreates the file `data/certificat_h_m11_J40674_100000.bin` of the table (same SHA-256) and keeps the compressed file. The uncompressed file is also deposited on Zenodo, in the record of this note.

Formats. A weights file contains 2^{40}, the number 13, then $p_1, \dots, p_{13}$. A certificate file contains, in little-endian 64-bit words, the number n of representatives, their n codes $\sum_i d_i 6^{m-i}$ (directions $0\colon +e_1$, $1\colon -e_1$, …, $5\colon -e_3$, d_1 the most recent step), the n values of H, then A and $B = 2^{30}$.

Commands (run in `supplement/`; `w.txt`, `c1.json`, `H.bin` are output files):

```
python3 certification/c1_poids_exacts.py 40674 100000 12 10 w.txt c1.json
python3 certification/c2_poids_flint.py 40674 100000 data/c1_J40674.json
gcc -O3 -fopenmp -o certification/tm_cert certification/tm_cert.c -lm
certification/tm_cert 11 w.txt H.bin
gcc -O3 -fopenmp -o certification/verif_cert certification/verif_cert.c
(cd data && xz -dk certificat_h_m11_J40674_100000.bin.xz)
certification/verif_cert 11 data/poids_m11_J40674_100000.txt \
                            data/certificat_h_m11_J40674_100000.bin
```

The first command recreates the weights file byte for byte; the second runs the Arb cross-check with the truncation guard (it needs a Python interpreter with python-flint installed); `tm_cert` recreates a certificate, and `c3_seuil.py` runs the dichotomy in J by calling it; `xz -dk` unpacks the supplied certificate; the last command prints `CERTIFICAT VALIDE` (“certificate valid”). Item 3 of Proposition 5.4 is the Python test

```
from fractions import Fraction; Fraction(40674,100000)*Fraction(2639854083,2**30) < 1
```

The folder `checks/` contains the programs and logs of the independent verifications of §§6.3 and 7. The runs used Python 3.14.4, python-flint 0.9.0, gcc 15.2.0 on x86-64. For $m = 13$, `verif_cert` took 2 min 36 s on 8 cores with 10.8 GB of resident memory, and the certificate was generated in about 20 minutes.

Formalisation in Lean. The folder `lean/` is a Lean 4 project (Lean 4.34.0, Mathlib v4.34.0) that states Theorem 2.1 for the Wilson measure on the torus $(\mathbb{Z}/L\mathbb{Z})^4$, with the normalised Haar measure on $SU(2) \cong S^3$, and proves part of its proof; the commands to rebuild it are in `supplement/README.md`. The statement is written for the rectangle with a corner at the origin

in the plane of the first and fourth axes (the other loops follow by symmetries of the torus, which are not formalised), and the expectation is the quotient of the two Haar integrals (the integrability of the integrands, which is clear for bounded continuous functions on a compact space, is not formalised). Its status is as follows.

- *Proved in the Lean kernel* (axioms `propext`, `Classical.choice` and `Quot.sound` only; no `sorry`, no `native_decide`; the main theorems and their closure, 43 458 declarations, were also accepted by the independent kernel nanoda): the combinatorial part of the proof, namely the Perron-type bound on the weighted walk counts, Lemmas 5.2 and 5.3, the step-by-step reading $\prod_z \widetilde{A}_{n_z} = \prod_j \widetilde{r}(N_j)$ of (4.3), the bound $|\omega| \geq R$ and the geometric series; the correctness of the certificate checker, for *every* function H; the certificates of memory $m = 3$, 4 and 5, by `decide +kernel` on the 6^m raw states (those of $m = 3, 4$ correspond to $\beta \leq 0.79718$, above $12/17$, and that of $m = 5$ to $\beta \leq 0.80438$); the final arithmetic in exact rationals ($\theta_* < 1$, $3.978 < K_0 < 3.979$, $563\,935 < C_* < 563\,936$); the elementary facts $\widetilde{A}_n > 0$ and $\widetilde{A}_1 \leq \frac{1}{2}$; and the junction theorem `theoreme1_de_verifiePar`: for every function H such that the checker returns `true` on the memory-11 data, and under the three hypotheses below, $|\langle W_\gamma \rangle| \leq C_*^T \theta_*^{RT}$ for $\beta \leq 0.81348$.

- *Established by compiled execution* (the trust is in the Lean and C compilers and in the runtime, not in the kernel): the memory-11 certificate, checked on all $6^{11} = 362\,797\,056$ raw states by the very function whose correctness is proved above. The values of H are read from the file of `tm_cert`; since the correctness holds for every H, a faulty reading could give `true` only if the values read were themselves a valid certificate.

- *Assumed*, as the three fields of the structure `HypothesesBS`: (i) the reduction $|\langle W_\gamma \rangle| \leq 2^T \widetilde{G}^T$ of (4.3), which gathers the slab decomposition, the walk representation, the trace formula, Lemma 4.1, Propositions 4.2 and 4.3 and Corollary 4.5 (§§3–4); (ii) the monotonicity of Lemma 5.1; (iii) the certified bounds on the weights, $\widetilde{r}(N; J_*) \leq p_{\min(N,13)}/2^{40}$ (Proposition 5.4(1); the rational enclosures of §6.1 are not redone in Lean).

Thus the Lean theorem for $\beta \leq 0.80438$ (memory 5, the three hypotheses taken at $J_* = 40219/10^5$ with the corresponding weights) rests on these three hypotheses alone, and the theorem for $\beta \leq 0.81348$ rests on them and on the compiled execution.

Version 1.1. The folder `supplement_v1.1/` contains the programs of §6.6, the weights files at $J = 0.405$, 0.40 and 0.35294 (the one at J_* is `data/poids_m11_J40674_100000.txt` of `supplement/`), the outputs of the runs, the re-check of the published certificate, and the SHA-256 of the vectors H (`data/SHA256_H.txt`); see its `README.md`. SHA-256 fingerprints:

`certification/cert_incline11.c`
`0dc2e7b0c881fee30d44e0bdbc942a0914e1afd4db520205e441b6cc38d3d0c1`

`certification/b7_verif11.py`
`38f063a2e49f2e6ef50ca3eb8f860607e818514434a16140e3c2f8d5992f8a07`

`certification/b7_constantes.py`
`c494a362d3e426203a3594848d70748076f962c37fd9ce1ab044671c7104f7c8`

`data/poids_m11_J40500_100000.txt`
`cc1eda638c4b6cce2c781ab96b067a712f3e9ba5b402480a4a00dd49071fc59c`

`data/poids_m11_J40000_100000.txt`
`cf8d8e869249f53512831b8df98c5051aad580912885d6f96bf271cd68ee6b83`

`data/poids_m11_J35294_100000.txt`
`17d23b03b618479f3f89cb2eab8757673edcd22454d779e177c1621dc9e0fab4`

`checks/table4_fractions.py`
`ccdc7486543c600dc591bd444be8706cbe867d2c4ad87fc6a0a00a9f64854678`

`data/SHA256_H.txt` (fingerprints of the six vectors H, 2 902 376 448 bytes each, on Zenodo)
`e0a76870a73d4745d4ae2f1fe02d661d68e081da8ac07f80631a365279436a39`

Commands (in `supplement_v1.1/`, after copying `poids_m11_J40674_100000.txt` from `supplement/data/` into `data/`; `H.bin` is an output file of 2.9 GB):

```
gcc -O3 -fopenmp -o certification/cert_incline11 certification/cert_incline11.c -lm
certification/cert_incline11 11 data/poids_m11_J40674_100000.txt 201 200 H.bin
python3 certification/b7_verif11.py 11 data/poids_m11_J40674_100000.txt \
        201 200 H.bin 2639868799
sha256sum H.bin
python3 certification/b7_constantes.py data data
python3 checks/table4_fractions.py data
```

The second command prints the certificate of the first row of Table 4 (zero violations), the third re-checks it, the fourth must give the fingerprint of `H_J40674_a201_b200.bin` in `data/SHA256_H.txt`, and the last two recompute the constants of Table 4 with two separate programs (the first of them needs python-flint). The other rows are obtained with the weights files and the integers a, b of the table. Instead of the second command, the vector can be unpacked from the file `H_J40674_a201_b200.bin.xz` of the Zenodo record (`xz -dk`), whose name then replaces `H.bin` in the next two commands.

Use of AI

The work reported here was carried out with AI models, under the author's direction.

- *Claude Opus 5.5 (Anthropic)* wrote the proof (§§3–5), the certification programs (`c1`–`c3`, `tm_cert`, `verif_cert`) and this note; in a separate session it carried out an adversarial verification with programs written independently of the certification programs (§6.3, §7).

- *Kimi K3.1 Pro (Moonshot AI)* wrote, independently, a program for the finite-memory operators with the weights of Borgs and Seiler, whose values coincide with ours; derived independently the inequality of Proposition 4.2, with a weaker site factorisation; carried out an adversarial review of the proof, with recompilation and re-execution of the programs; audited the memory-13 certificate with its own program (§6.3); and reviewed a previous version of this note (before the journal page numbers of [2] and the references [17, 11] were added), regenerating the weights file and the memory-11 certificate byte for byte (no blocking defect; two minor corrections applied); it also reviewed the statement of the Lean formalisation, the boundary between what is proved, executed and assumed, and the arbitration of the objections (two corrections applied).

- *DeepSeek V4 Pro (DeepSeek)* contributed exploratory derivations of other approaches and four adversarial reviews of the proof; their objections led to corrections of the write-up (the orientation of the loop C_T in Proposition 3.5, the comparison of exact rather than rounded weights in §5.4, and the constant K_0). In a last flaw hunt, ten independent adversarial instances (one per step of the proof) and seven cross-critiques raised eight objections (one of them withdrawn by its author); the arbitration, by derivation or computation, found no mathematical flaw (the two objections called fatal were refuted by an exact Haar integral and by a counting argument) and led to three changes of the write-up: the range $0 < J \leq 1$ of Lemma 4.1, the statement on the string tension (Corollary 2.2) and the remark on the range of R (Remark 2.4).

- *Claude (Anthropic)*, in later sessions, wrote the Lean formalisation of §9 and arbitrated the objections of the flaw hunt.

- *Version 1.1.* Claude Opus 5.5 (Anthropic) wrote §5.5 and §6.6, the three programs of §6.6 (`cert_incline11.c`, `b7_verif11.py` and `b7_constantes.py`) and this revision; the use of a tilt and of the parity for the constants was also proposed, independently, by

DeepSeek V4 Pro in an exploratory session. Kimi K3.1 Pro reviewed an internal write-up of the tilted bound and its programs (recompilation, re-execution at memory 7, negative control). The draft of version 1.1 was reviewed adversarially by DeepSeek V4 Pro (two reviews) and, Kimi being unavailable (quota), by a separate instance of Claude Opus 5.5 reading the PDF without prior context, which also wrote `checks/table4_fractions.py`; see the section on the changes from version 1.0.

These reviews are not a substitute for refereeing: no human has checked the proof.

Changes from version 1.0

- New §5.5: the tilted bound (Theorem 5.6), its memory-11 certificates (Proposition 5.7, Table 4), Corollary 5.8 and Remark 5.9. At $\beta \leq 0.81348$ the certified lower bound on the string tension rises from $7.05 \cdot 10^{-6}$ to 0.004989, and the radius beyond which the bound is non-trivial on every torus drops from $1.9 \cdot 10^{6}$ to $3.1 \cdot 10^{3}$; for $\beta \leq 0.80$, from 0.0133 to 0.2253 and from 430 to 36.5. The threshold, Theorem 2.1, its certificate and the Lean formalisation are unchanged; Theorem 5.6 is not formalised.

- New §6.6 (programs and verification of the tilted certificates, Remark 6.1 on the two certificates at $\eta = 0$) and folder `supplement_v1.1/` (§9); the six vectors H of the tilted certificates are deposited, compressed with `xz`, in the Zenodo record of this version. The files of version 1.0 are unchanged.

- Abstract, introduction (§§1.2 and 1.4), Remark 2.4, the caption of Table 2, §8.1 and the statement on the use of AI updated accordingly. The new Table 4 shifts the former Table 4 to Table 5; section numbers are unchanged.

- Review of the draft of version 1.1: two adversarial reviews by DeepSeek V4 Pro (the two objections called blocking were a request for more explicit wording, now applied, and an arithmetic slip; the other objections were applied or refuted by computation) and a reading of the PDF without prior context by a separate instance of Claude Opus 5.5 (Kimi K3.1 Pro being unavailable), which found no mathematical flaw and recomputed Table 4 with its own program; its corrections are included (comparison with [16, Cor. 7.2] in Remark 2.4, rounding in the caption of Table 4, radii on every torus, scope of Remark 5.9, controls at the unit level and Remark 6.1). No human has refereed version 1.1.

References

[1] S. E. Alm, *Upper bounds for the connective constant of self-avoiding walks*, Combin. Probab. Comput. 2 (1993) 115–136, doi:10.1017/S0963548300000547.

[2] C. Borgs, E. Seiler, *Lattice Yang–Mills theory at nonzero temperature and the confinement problem*, Comm. Math. Phys. 91 (1983) 329–380, doi:10.1007/BF01208780 (page numbers refer to this version); preprint MPI-PAE/PTh 16/83 (March 1983), with the same text and equation numbers in the passages cited.

[3] M. Caselle, M. Hasenbusch, *The stability of the $O(N)$ invariant fixed point in three dimensions*, J. Phys. A 31 (1998) 4603–4617, doi:10.1088/0305-4470/31/20/004, arXiv:cond-mat/9711080.

[4] S. Cao, R. Nissim, S. Sheffield, *Dynamical approach to area law for lattice Yang–Mills*, arXiv:2509.04688 (2025).

[5] R. G. Douglas, *On majorization, factorization, and range inclusion of operators on Hilbert space*, Proc. Amer. Math. Soc. 17 (1966) 413–415, doi:10.1090/S0002-9939-1966-0203464-1.

[6] B. Durhuus, J. Fröhlich, *A connection between ν-dimensional Yang–Mills theory and $(\nu-1)$-dimensional, non-linear σ-models*, Comm. Math. Phys. 75 (1980) 103–151, doi:10.1007/BF01222514; preprint IHES/P/79/302 (1979), to which "preprint p." refers.

[7] A. M. Ferrenberg, J. Xu, D. P. Landau, *Pushing the limits of Monte Carlo simulations for the three-dimensional Ising model*, Phys. Rev. E 97 (2018) 043301, doi:10.1103/PhysRevE.97.043301.

[8] J. Fröhlich, *Confinement in Z_n lattice gauge theories implies confinement in $SU(n)$ lattice Higgs theories*, Phys. Lett. B 83 (1979) 195–198, doi:10.1016/0370-2693(79)90684-1.

[9] J. Fröhlich, *Lectures on Yang–Mills theory*, preprint IHES/P/80/01, Institut des Hautes Études Scientifiques (1980).

[10] J. Fröhlich, *Some comments on the crossover between strong and weak coupling in $SU(2)$ pure Yang–Mills theory*, Phys. Rep. 67 (1980) 137–149, doi:10.1016/0370-1573(80)90086-1; preprint IHES/P/80/20 (Les Houches lectures, February 1980), to which page and equation numbers refer.

[11] K. R. Ito, E. Seiler, *Further discussion of Tomboulis' approach to the confinement problem*, arXiv:0803.3019 (2008).

[12] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017) 1281–1292, doi:10.1109/TC.2017.2690633.

[13] K. Kanaya, S. Kaya, *Critical exponents of a three-dimensional $O(4)$ spin model*, Phys. Rev. D 51 (1995) 2404–2410, doi:10.1103/PhysRevD.51.2404, arXiv:hep-lat/9409001.

[14] K. Osterwalder, E. Seiler, *Gauge field theories on a lattice*, Ann. Phys. 110 (1978) 440–471, doi:10.1016/0003-4916(78)90039-8.

[15] A. Pönitz, P. Tittmann, *Improved upper bounds for self-avoiding walks in $\mathbb{Z}^d$*, Electron. J. Combin. 7 (2000) R21, doi:10.37236/1499.

[16] K. Rémondière, *Dobrushin uniqueness for the four-dimensional $SU(2)$ Wilson lattice gauge measure for $\beta < 2/9$, with interval/Lean certificates*, note v1.5 (1 November 2026), Zenodo, doi:10.5281/zenodo.23856414; Section 7: area law for $\beta < 2/3$.

[17] E. T. Tomboulis, *Confinement for all values of the coupling in four-dimensional $SU(2)$ gauge theory*, arXiv:0707.2179 (2007).

[18] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D 10 (1974) 2445–2459, doi:10.1103/PhysRevD.10.2445.